

Stochastic Processes

Throughout, $(\Omega, \mathcal{F}, \mathbb{P})$ is one fixed probability space. The lecture is organised around a single picture, a **table of sample paths**: one row for each outcome ω , one column for each time t , and the value $X_t(\omega)$ in the cell.

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|------------------------------|---|
| 1. Stochastic process | A family of random variables indexed by time. Rows are paths, columns are random variables. |
| 2. Random walk | Sum of independent ± 1 steps. The reflection principle and first-step analysis. |
| 3. Brownian motion | The limit of the random walk under $\sqrt{n}$ scaling; independent Gaussian increments. |
| 4. Risk of ruin | The gambler's ruin problem, solved by first-step analysis. |
| 5. Markov chains | The transition matrix is a table whose rows are conditional distributions. Powers, stationary distributions, classes, hitting times, convergence. |

Stochastic Process: Definition

Definition

Let T be a set of **times**, either $T = \{0, 1, 2, \dots\}$ (discrete time) or $T = [0, \infty)$ (continuous time). A **stochastic process** is a family

$$(X_t)_{t \in T}, \quad X_t: \Omega \rightarrow \mathbb{R} \text{ a random variable for each } t \in T,$$

all defined on the same $(\Omega, \mathcal{F}, \mathbb{P})$. Equivalently it is one function of two arguments, $X: T \times \Omega \rightarrow \mathbb{R}$, $(t, \omega) \mapsto X_t(\omega)$.

There is one random variable for each time, and all are defined on the same Ω .

- **For fixed** ω , the function $T \rightarrow \mathbb{R}$, $t \mapsto X_t(\omega)$, is the **sample path** (or trajectory) of the outcome ω : an ordinary sequence or an ordinary function of time.
- **For fixed** t , the function $\Omega \rightarrow \mathbb{R}$, $\omega \mapsto X_t(\omega)$, is the random variable X_t : what the process shows at time t .
- The **state space** S is the set of values the X_t take, so $X_t: \Omega \rightarrow S$: usually $S \subseteq \mathbb{R}$ ($S = \{-1, +1\}, \mathbb{Z}, \mathbb{R}$); for Markov chains a finite or countable set of named states.
- Shorthand, used throughout: for $A \subseteq S$, $\{X_t \in A\}$ is the event $\{\omega \in \Omega : X_t(\omega) \in A\}$ and $\mathbb{P}(X_t \in A) = \mathbb{P}(\{\omega \in \Omega : X_t(\omega) \in A\})$; $\mathbb{P}(X_s = x, X_t = y)$ is the probability of the intersection of the two such events.

The Central Picture: A Table of Sample Paths

Three fair coin tosses. $\Omega = \{H, T\}^3$, each ω having probability $\frac{1}{8}$. Put $\xi_t: \Omega \rightarrow \{-1, +1\}$, $\xi_t(\omega) = +1$ if toss t of ω is H and -1 if T, and

$$S_t = \xi_1 + \cdots + \xi_t, \quad S_0 = 0, \quad t \in \{0, 1, 2, 3\}.$$

- A **row** is one outcome ω and shows the path $t \mapsto S_t(\omega)$.
- A **column** is one time t and shows the random variable S_t : its entries, with the row probabilities, are the distribution of S_t .
- An event is a set of rows: $\{S_3 = 1\}$ is rows 2, 3, 5, so $\mathbb{P}(S_3 = 1) = \frac{3}{8}$.

Properties of the paths are read along the rows of this table, and properties of the distribution at a fixed time are read down the columns.

ω	$\mathbb{P}(\{\omega\})$	S_0	S_1	S_2	S_3
HHH	$\frac{1}{8}$	0	1	2	3
HHT	$\frac{1}{8}$	0	1	2	1
HTH	$\frac{1}{8}$	0	1	0	1
HTT	$\frac{1}{8}$	0	1	0	-1
THH	$\frac{1}{8}$	0	-1	0	1
THT	$\frac{1}{8}$	0	-1	0	-1
TTH	$\frac{1}{8}$	0	-1	-2	-1
TTT	$\frac{1}{8}$	0	-1	-2	-3

rows: paths $t \mapsto S_t(\omega)$; columns: random variables S_t

Reading the Table Two Ways, and What a Process Is Not

Row questions (about paths)

Whether the path reaches 2, whether it returns to 0, and what its maximum is. Each is a set of rows, hence an event: $\{\max_{t \leq 3} S_t \geq 2\}$ is rows 1, 2, probability $\frac{2}{8}$.

Column questions (about one time)

The values of $\mathbb{E}[S_2]$ and $\mathbb{P}(S_3 > 0)$. Each is read from column 2 or 3 together with the row probabilities:

$$\mathbb{E}[S_2] = \frac{1}{8}(2 + 2 + 0 + 0 + 0 + 0 - 2 - 2) = 0,$$
$$\mathbb{P}(S_3 > 0) = \frac{4}{8}.$$

Mixed questions (about several columns)

$\mathbb{P}(S_1 = 1, S_3 = 1)$ is rows 2, 3: $\frac{2}{8}$. Different columns are *not* independent in general:

$$\mathbb{P}(S_1 = 1)\mathbb{P}(S_3 = 1) = \frac{1}{2} \cdot \frac{3}{8} \neq \frac{2}{8}.$$

A stochastic process is **not**:

- a single random variable: it is one per time;
- a list of distributions, one per column: the distributions of the S_t separately do not determine how columns move together, i.e. the joint distribution of the row;
- a fixed function of time: the path depends on ω , and different rows are different functions.

The process is the entire table; its columns are the random variables X_t and its rows are the possible histories.

The Distribution of a Process

Definition

The **finite-dimensional distributions** of $(X_t)_{t \in T}$ are the joint distributions of $(X_{t_1}, \dots, X_{t_n})$ for all n and all times $t_1 < \dots < t_n$ in T : in the discrete case the joint pmfs

$$p_{X_{t_1}, \dots, X_{t_n}} : \mathbb{R}^n \rightarrow [0, 1], \quad (x_1, \dots, x_n) \mapsto \mathbb{P}(X_{t_1} = x_1, \dots, X_{t_n} = x_n).$$

The model specifies the joint distribution of every finite set of columns, and these joint distributions together determine every probability concerning the process.

Definition (independence)

Random variables $X_1, \dots, X_n$ are **independent**, written $X_1 \perp\!\!\!\perp X_2 \perp\!\!\!\perp \dots \perp\!\!\!\perp X_n$, if for all sets $A_1, \dots, A_n \subseteq \mathbb{R}$

$$\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) = \mathbb{P}(X_1 \in A_1) \cdots \mathbb{P}(X_n \in A_n).$$

A sequence $(X_n)_{n \geq 1}$ is independent if every finite sub-family is.

- In the table, the coin tosses ξ_1, ξ_2, ξ_3 are independent, but the partial sums S_1, S_2, S_3 are not (previous slide). In practice a model is specified by independent building blocks together with a rule for combining them, rather than by listing all the joint pmfs.

The Bernoulli Process and the Simple Random Walk

Definition

Fix $p \in (0, 1)$ and put $q = 1 - p$. A **Bernoulli process** is an independent sequence $\xi_1, \xi_2, \dots$ of random variables $\xi_n: \Omega \rightarrow \{-1, +1\}$ with $\mathbb{P}(\xi_n = +1) = p$, $\mathbb{P}(\xi_n = -1) = q$ for every n . The **simple random walk** is the process $(S_n)_{n \geq 0}$,

$$S_n: \Omega \rightarrow \mathbb{Z}, \quad S_0 = 0, \quad S_n = \xi_1 + \xi_2 + \dots + \xi_n.$$

It is **symmetric** (or fair) if $p = \frac{1}{2}$.

A coin is tossed repeatedly and a running score is kept, +1 for each head and -1 for each tail; S_n is the score after n tosses.

- The table on slide 3 is this process up to $n = 3$. Each row is a path $n \mapsto S_n(\omega)$ that moves by ± 1 at each step.
- **Independent increments:** for $m < n$, $S_n - S_m = \xi_{m+1} + \dots + \xi_n$ depends only on tosses after m , so it is independent of $(S_0, \dots, S_m)$ and has the same distribution as S_{n-m} . After time m , the process $(S_{m+k} - S_m)_{k \geq 0}$ is a simple random walk independent of $(S_0, \dots, S_m)$.

Distribution of S_n

Theorem

Let $H_n: \Omega \rightarrow \{0, \dots, n\}$, $H_n = \sum_{k=1}^n \mathbf{1}\{\xi_k = +1\}$, be the number of $+1$ steps among the first n : a sum of n independent Bernoulli(p) indicators, hence binomial(n, p), and $S_n = H_n - (n - H_n) = 2H_n - n$. Hence, for $k \in \{-n, -n+2, \dots, n\}$,

$$\mathbb{P}(S_n = k) = \binom{n}{\frac{n+k}{2}} p^{(n+k)/2} q^{(n-k)/2}, \quad \mathbb{E}[S_n] = n(p - q), \quad \text{Var}(S_n) = 4npq.$$

S_n is an affine function of a binomial random variable; its mean is of order n and its standard deviation of order $\sqrt{n}$.

Proof

$S_n = k$ iff $H_n = \frac{n+k}{2}$, which requires $n+k$ even. $\mathbb{E}[S_n] = 2\mathbb{E}[H_n] - n = 2np - n = n(p - q)$ and $\text{Var}(S_n) = 4 \text{Var}(H_n) = 4npq$. $\square$

Fair walk, $n = 4$

k	-4	-2	0	2	4
$\mathbb{P}(S_4 = k)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

$\mathbb{E}[S_4] = 0$, $\text{Var}(S_4) = 4$, standard deviation $2 = \sqrt{4}$.

A Path Question: Reaching 3 Within 10 Steps

For the fair walk, consider $\mathbb{P}(\max_{k \leq 10} S_k \geq 3)$. This is a row question, since the event depends on the entire path rather than on the value at a single time.

Direct computation

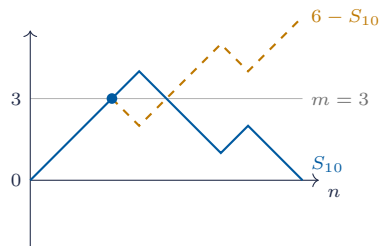
One can enumerate the $2^{10} = 1024$ paths and count those whose running maximum reaches 3. The following theorem replaces this count by a computation involving the distribution of S_n only.

Theorem (reflection principle, fair walk)

For integers $m \geq 1$ and $n \geq 1$,

$$\mathbb{P}\left(\max_{k \leq n} S_k \geq m\right) = 2\mathbb{P}(S_n > m) + \mathbb{P}(S_n = m).$$

The distribution of the maximum up to time n is expressed through the distribution of S_n .



After the first visit to m , reflect the rest of the path in the line m . This pairs each path that reaches m and ends below m with one ending above m , and the two have equal probability.

Reflection Principle: Proof

Proof

Let $A = \{\max_{k \leq n} S_k \geq m\}$. Split A by where the path ends:

$$\mathbb{P}(A) = \mathbb{P}(A, S_n > m) + \mathbb{P}(A, S_n = m) + \mathbb{P}(A, S_n < m).$$

In the first two terms the condition A is redundant, since a path ending at or above m has reached m ; they equal $\mathbb{P}(S_n > m)$ and $\mathbb{P}(S_n = m)$.

For the third, let τ be the first time the path hits m . Map each path in $\{A, S_n < m\}$ to the path that agrees with it up to time τ and thereafter takes the opposite steps. The new path ends at $m + (m - S_n) > m$, and it has the same number of steps, each of probability $\frac{1}{2}$, so the same probability 2^{-n} . The map is a bijection onto $\{S_n > m\}$ (reverse it by reflecting after the first visit to m). Hence $\mathbb{P}(A, S_n < m) = \mathbb{P}(S_n > m)$. $\square$

For the fair walk all 2^n paths are equally likely, so probabilities are proportional to counts, and reflection is a bijection between two sets of paths.

- For $p \neq \frac{1}{2}$ the reflected path has a different probability, and the identity fails; the argument then yields only inequalities.
- Brownian motion inherits the statement in the form $\mathbb{P}(\max_{s \leq t} B_s \geq a) = 2\mathbb{P}(B_t \geq a)$ (slide 23).

Return to the Problem: Reaching 3 Within 10 Steps

The answer

S_{10} is even, so $\mathbb{P}(S_{10} = 3) = 0$, and

$$\mathbb{P}(S_{10} > 3) = \mathbb{P}(S_{10} \geq 4) = \mathbb{P}(H_{10} \geq 7) = \frac{120 + 45 + 10 + 1}{1024} = \frac{11}{64}. \text{ Hence}$$

$$\mathbb{P}\left(\max_{k \leq 10} S_k \geq 3\right) = 2 \cdot \frac{11}{64} + 0 = \frac{11}{32} \approx 0.34.$$

Direct enumeration of the 1024 paths gives 352, in agreement.

Corollary: the distribution of the maximum

With $M_n = \max_{k \leq n} S_k$ and the theorem applied at m and $m + 1$,

$$\mathbb{P}(M_n = m) = \mathbb{P}(M_n \geq m) - \mathbb{P}(M_n \geq m + 1) = \mathbb{P}(S_n = m) + \mathbb{P}(S_n = m + 1) \quad (m \geq 0).$$

The distribution of M_n , a path quantity, is determined by the distribution of S_n , a single column.

- Fair walk, $n = 10$: $\mathbb{P}(M_{10} = 0) = \mathbb{P}(S_{10} = 0) + \mathbb{P}(S_{10} = 1) = \frac{252}{1024}$, so about a quarter of all 10-step paths never rise above their start.

First-Step Analysis: Expected Time to Leave $(-a, a)$

Fair walk started at $i \in \{-a+1, \dots, a-1\}$, position $X_n = i + S_n$. Write $\mathbb{P}_i(\cdot)$, $\mathbb{E}_i[\cdot]$ for probabilities and expectations when the start is i (the same notation is used for Markov chains below). **Exit time:** $\tau: \Omega \rightarrow \{0, 1, \dots\} \cup \{\infty\}$, $\tau = \min\{n \geq 0 : |X_n| \geq a\}$ (∞ if there is no such n); $m_i = \mathbb{E}_i[\tau]$.

Direct computation

$\mathbb{P}_i(\tau = n)$ is found by counting the paths that stay strictly inside $(-a, a)$ for $n-1$ steps and exit at step n ; then $\sum_n n \mathbb{P}_i(\tau = n)$. These counts have no simple closed form.

First-step analysis

Condition on the first step, as for the waiting time for HH (Conditional Expectation lecture, Application 2). After one step the walk is at $i \pm 1$ and, by independent increments, the remaining steps $\xi_2, \xi_3, \dots$ form a simple random walk independent of ξ_1 , so the expected exit time from that point is $m_{i \pm 1}$:

$$m_i = 1 + \frac{1}{2}m_{i+1} + \frac{1}{2}m_{i-1} \quad (-a < i < a), \quad m_{-a} = m_a = 0.$$

The expected exit time from i is one step plus the average of the expected exit times from the two neighbouring positions.

- The unknowns $m_{-a+1}, \dots, m_{a-1}$ appear on both sides, so this is a system of $2a-1$ linear equations rather than an explicit formula; every first-step argument in this lecture has this form.

First-Step Analysis: The Solution

Theorem

$m_i = \mathbb{E}_i[\tau] = a^2 - i^2$ for $-a \leq i \leq a$; in particular $\mathbb{E}_0[\tau] = a^2$.

Proof

Try $m_i = c - i^2$: then $1 + \frac{1}{2}(c - (i+1)^2) + \frac{1}{2}(c - (i-1)^2) = 1 + c - i^2 - 1 = c - i^2$, so the equation holds for every c , and $m_{\pm a} = 0$ forces $c = a^2$. The system is $2a - 1$ linear equations in $2a - 1$ unknowns; it has a unique solution (uniqueness: if m, m' both solve it, $d = m - m'$ satisfies $d_i = \frac{1}{2}(d_{i+1} + d_{i-1})$ with $d_{\pm a} = 0$, so d is linear and vanishes). $\square$

The expected exit time is a quadratic function of the starting point, and from the centre it equals a^2 .

$a = 3$

i	-3	-2	-1	0	1	2	3
m_i	0	5	8	9	8	5	0

From 0 the walk takes 9 steps on average to reach ± 3 ; from ± 2 it takes 5.

This is consistent with slide 7: the walk travels a distance of order $\sqrt{n}$ in n steps, so a distance a takes a number of steps of order a^2 .

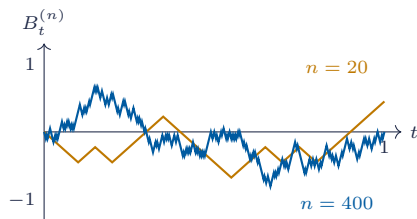
From the Walk to Brownian Motion: The $\sqrt{n}$ Scaling

A particle takes a fair ± 1 step every $\frac{1}{n}$ seconds, each of size $\frac{1}{\sqrt{n}}$. We ask what its path looks like when n is large, and with what probability it is above 0 at time $t = 1$ and again at time $t = 2$.

Rescaling

Position at time t is $B_t^{(n)} = \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor}$. By slide 7, $\mathbb{E}[B_t^{(n)}] = 0$ and $\text{Var}(B_t^{(n)}) = \frac{\lfloor nt \rfloor}{n} \rightarrow t$. The central limit theorem (CLT) states: for i.i.d. ξ_k with mean 0 and variance 1, $\mathbb{P}(S_m/\sqrt{m} \leq x) \rightarrow \Phi(x)$ for every x , with $\Phi: \mathbb{R} \rightarrow [0, 1]$ the $N(0, 1)$ cdf. Applied with $m = \lfloor nt \rfloor$ it gives $\mathbb{P}(B_t^{(n)} \leq x) \rightarrow \Phi(x/\sqrt{t})$: $B_t^{(n)} \rightarrow N(0, t)$ in distribution. Increments over disjoint time intervals are sums of disjoint blocks of steps, so they are independent, with variance equal to the length of the interval.

Division by $\sqrt{n}$ is the scaling under which the variance at time t has a finite non-zero limit; the limiting process has normal marginals and independent increments.



The same coin sequence, 20 and 400 steps, each rescaled to fit $[0, 1]$ with steps of size $1/\sqrt{n}$. At time 1 both have variance 1.

The Invariance Principle

Theorem (invariance principle, Donsker 1951; informal statement)

Let $\xi_1, \xi_2, \dots$ be independent with $\mathbb{E}[\xi_k] = 0$ and $\text{Var}(\xi_k) = 1$, $S_n = \xi_1 + \dots + \xi_n$, and $B_t^{(n)} = \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor}$. Then, as $n \rightarrow \infty$, the random path $t \mapsto B_t^{(n)}$ converges in distribution to a random continuous path $t \mapsto B_t$, and for all times $t_1 < \dots < t_k$ and sets $A_1, \dots, A_k$,

$$\mathbb{P}(B_{t_1}^{(n)} \in A_1, \dots, B_{t_k}^{(n)} \in A_k) \rightarrow \mathbb{P}(B_{t_1} \in A_1, \dots, B_{t_k} \in A_k).$$

The limit process B is the same for every step distribution with mean 0 and variance 1.

On a large scale, every random walk with mean-zero, unit-variance steps has approximately the same distribution of paths, and the limiting distribution is that of Brownian motion.

- The theorem is not proved here. The name refers to the fact that the limit does not depend on the step distribution; the central limit theorem is the special case of a single time $t_1 = 1$.
- The limit inherits three properties from the walk: independent increments over disjoint intervals (sums over disjoint blocks of steps), $\text{Var}(\text{increment}) = \text{length}$, and a start at 0. Since the step size $1/\sqrt{n}$ tends to 0, the limiting paths are continuous. These are the four properties in the definition on the next slide.

Brownian Motion: Definition

Definition

A **(standard) Brownian motion** is a process $B: [0, \infty) \times \Omega \rightarrow \mathbb{R}$, $(t, \omega) \mapsto B_t(\omega)$, such that

- (i) $B_0 = 0$;
- (ii) **independent increments:** for all $0 \leq t_0 < t_1 < \dots < t_n$, the random variables $B_{t_1} - B_{t_0}, B_{t_2} - B_{t_1}, \dots, B_{t_n} - B_{t_{n-1}}$ are independent;
- (iii) **Gaussian increments:** for $0 \leq s < t$, $B_t - B_s \sim N(0, t - s)$, i.e. it has pdf
$$x \mapsto \frac{1}{\sqrt{2\pi(t-s)}} e^{-x^2/(2(t-s))};$$
- (iv) **continuous paths:** for every ω , the function $[0, \infty) \rightarrow \mathbb{R}$, $t \mapsto B_t(\omega)$, is continuous.

Over any time interval the displacement is normal with variance equal to the length of the interval, displacements over disjoint intervals are independent, and the paths have no jumps.

- Existence is a theorem (Wiener, 1923); we take it as given. The four properties determine all finite-dimensional distributions (slide 5), hence every probability about B .

What B Is: Reading the Table

The table of paths now has a continuum of columns ($t \geq 0$) and of rows (ω); the two readings are unchanged.

Column t : the random variable $B_t: \Omega \rightarrow \mathbb{R}$

$B_t = B_t - B_0 \sim N(0, t)$ by (i) and (iii). Its pdf is $f_{B_t}: \mathbb{R} \rightarrow [0, \infty)$, $f_{B_t}(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}$, so $\mathbb{P}(a \leq B_t \leq b) = \int_a^b f_{B_t}(x) dx$.

Row ω : the path $[0, \infty) \rightarrow \mathbb{R}$, $t \mapsto B_t(\omega)$

A continuous function starting at 0. Different rows are different functions; a typical one is as irregular as the $n = 400$ picture at every magnification. Row questions include the maximum up to time t , the first time the path reaches a , and the time spent above 0.

The two arguments t and ω

In $B_t(\omega)$, t is the time (which column) and ω is the outcome (which row). For fixed ω , $t \mapsto B_t(\omega)$ is a function of time; for fixed t , $\omega \mapsto B_t(\omega)$ is a random variable. An increment $B_t - B_s$ is a difference of two columns.

The properties below all follow from writing values of B as sums of independent increments.

What B Is Not

Brownian motion is **not**:

- a process with independent *values*: B_s and B_t are strongly dependent (next slide); it is the *increments* over disjoint intervals that are independent;
- stationary in value: $\text{Var}(B_t) = t$ grows; it is the increments whose distribution depends only on the interval length;
- smooth: no path has a derivative anywhere (slide 24);
- a single normal random variable: it is one normal per t , and jointly they form a Gaussian family.

B has the same increment structure as the walk, in continuous time and with continuous state space.

	walk S_n	Brownian B_t
time set T	$\{0, 1, 2, \dots\}$	$[0, \infty)$
state space	$\mathbb{Z}$	$\mathbb{R}$
start	$S_0 = 0$	$B_0 = 0$
increment over $(s, t]$	sum of $t - s$ steps	$N(0, t - s)$
Var of increment	$t - s$ (fair)	$t - s$
disjoint increments	independent	independent
path	moves by ± 1	continuous, nowhere differentiable

Basic Properties: Marginals and Covariance

Theorem

- (a) $B_t \sim N(0, t)$: $\mathbb{E}[B_t] = 0$, $\text{Var}(B_t) = t$.
- (b) $\text{Cov}(B_s, B_t) = \mathbb{E}[B_s B_t] = \min(s, t)$ for all $s, t \geq 0$.

Each B_t is $N(0, t)$, and the covariance of two values is the earlier of the two times.

Proof

(a) is (i) and (iii) with $s = 0$. (b) For $s < t$ write $B_t = B_s + (B_t - B_s)$. The increment is independent of B_s (property (ii) applied to $0 < s < t$) and has mean 0, so

$$\mathbb{E}[B_s B_t] = \mathbb{E}[B_s^2] + \mathbb{E}[B_s (B_t - B_s)] = s + \mathbb{E}[B_s] \mathbb{E}[B_t - B_s] = s + 0 = \min(s, t).$$

Since $\mathbb{E}[B_s] = 0$, $\text{Cov}(B_s, B_t) = \mathbb{E}[B_s B_t]$. □

- The correlation of B_s and B_t is $\min(s, t)/\sqrt{st} = \sqrt{s/t}$ for $s < t$: B_1 and B_2 have correlation $1/\sqrt{2} \approx 0.71$; B_1 and B_{100} have correlation 0.1. The values are dependent although the increments are independent.
- Alternatively, $\mathbb{E}[B_s B_t]$ is a two-dimensional integral against the joint pdf of the next slide; the proof above avoids it.

Basic Properties: Joint Probability Density Function (pdf) of Several Columns

Theorem

For $0 = t_0 < t_1 < \dots < t_n$, the vector $(B_{t_1}, \dots, B_{t_n})$ has joint pdf $f: \mathbb{R}^n \rightarrow [0, \infty)$,

$$f(x_1, \dots, x_n) = \prod_{k=1}^n \frac{1}{\sqrt{2\pi(t_k - t_{k-1})}} \exp\left(-\frac{(x_k - x_{k-1})^2}{2(t_k - t_{k-1})}\right), \quad x_0 = 0.$$

Any finite collection of values of B has a joint density, obtained from the product of the increment densities by the substitution $d_k = x_k - x_{k-1}$.

Proof

The increments $D_k = B_{t_k} - B_{t_{k-1}}$, $k = 1, \dots, n$, are independent $N(0, t_k - t_{k-1})$ by (ii) and (iii), so $(D_1, \dots, D_n)$ has the product pdf $\prod_k \frac{1}{\sqrt{2\pi(t_k - t_{k-1})}} e^{-d_k^2/(2(t_k - t_{k-1}))}$. The values are

$B_{t_k} = D_1 + \dots + D_k$: a linear map from $(d_1, \dots, d_n)$ to $(x_1, \dots, x_n)$ with inverse $d_k = x_k - x_{k-1}$, of determinant 1. A determinant-one change of variables leaves a pdf unchanged apart from the substitution. □

Basic Properties: Scaling, Symmetry, Restart

Theorem

Let B be a Brownian motion. Each of the following is again a Brownian motion:

- (a) **Scaling:** $\left(\frac{1}{\sqrt{c}} B_{ct}\right)_{t \geq 0}$ for any fixed $c > 0$.
- (b) **Symmetry:** $(-B_t)_{t \geq 0}$.
- (c) **Restart:** $(B_{s+t} - B_s)_{t \geq 0}$ for any fixed $s \geq 0$, and it is independent of $(B_u)_{0 \leq u \leq s}$.

Rescaling, reflecting in the time axis, and restarting from a fixed time each give a process with the same distribution as B .

Proof

Check (i)–(iv). (a) $\frac{1}{\sqrt{c}} B_{c \cdot 0} = 0$; increments $\frac{1}{\sqrt{c}}(B_{ct} - B_{cs})$ over disjoint intervals are independent, normal with variance $\frac{1}{c}(ct - cs) = t - s$; paths are continuous. (b) $-N(0, \sigma^2)$ is $N(0, \sigma^2)$. (c) Increments of the new process are increments of B over intervals inside $[s, \infty)$, independent of increments over intervals inside $[0, s]$, and B_u for $u \leq s$ is such an increment. □

By (a) with $c = n$, $\frac{1}{\sqrt{n}} B_{nt}$ is again Brownian: the scaling of slide 13 applied to B itself. (c) is the continuous-time form of the restart property used in first-step analysis: after any fixed time s , $B_{s+t} - B_s$ is a Brownian motion independent of the past.

Return to the Problem: $\mathbb{P}(B_2 > 0 \mid B_1 = 1)$

By the independent increment

$B_2 = B_1 + (B_2 - B_1)$, and $B_2 - B_1 \sim N(0, 1)$ is independent of B_1 . On the row $\{B_1 = 1\}$, $B_2 = 1 + (B_2 - B_1)$, so given $B_1 = 1$, $B_2 \sim N(1, 1)$ and

$$\mathbb{P}(B_2 > 0 \mid B_1 = 1) = \mathbb{P}(N(1, 1) > 0) = \mathbb{P}(N(0, 1) > -1) = \Phi(1) \approx 0.841,$$

with Φ the $N(0, 1)$ cdf of slide 13.

Via the joint pdf

By slide 19 with $t_1 = 1$, $t_2 = 2$, (B_1, B_2) has joint pdf $f(x, y) = \frac{1}{2\pi} e^{-x^2/2} e^{-(y-x)^2/2}$, a bivariate normal with covariance matrix $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$. Form $f_{B_2|B_1}(y \mid x) = f(x, y)/f_{B_1}(x) = \frac{1}{\sqrt{2\pi}} e^{-(y-x)^2/2}$ and integrate over $y > 0$ at $x = 1$. The conditional density is that of $N(x, 1)$, and the answer is the same.

Conditioning on B_1 fixes the first increment and does not affect the distribution of the second; conditional probabilities for B are computed by splitting values into increments.

- The same split gives the conditional distribution of B_t given $B_s = x$ for any $s < t$: $N(x, t - s)$. The conditional distribution depends on the past only through the present value.

Return to the Opening Problem: $\mathbb{P}(B_1 > 0, B_2 > 0)$

Let $X = B_1$ and $Z = B_2 - B_1$: independent, both $N(0, 1)$, and $B_2 = X + Z$. The event is $\{X > 0, X + Z > 0\}$.

By the independent increment and symmetry

Split according to the sign of Z :

$\{X > 0, X + Z > 0\} = \{X > 0, Z > 0\} \cup \{X > 0, Z < 0, X > -Z\}$. The first event has probability $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ by independence. For the second: on the row $\{X > 0, Z < 0\}$, which has probability $\frac{1}{4}$, the pair $(X, -Z)$ consists of two independent positive copies of the same distribution, so $X > -Z$ has conditional probability $\frac{1}{2}$ by symmetry. Hence

$$\mathbb{P}(B_1 > 0, B_2 > 0) = \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{2} = \frac{3}{8}.$$

Via the joint pdf

Alternatively, integrate the joint pdf $\frac{1}{2\pi} e^{-x^2/2} e^{-(y-x)^2/2}$ over the quadrant $x > 0, y > 0$. This is the orthant probability of a bivariate normal with correlation $\rho = 1/\sqrt{2}$, which by a polar-coordinate computation equals $\frac{1}{4} + \frac{\arcsin \rho}{2\pi} = \frac{1}{4} + \frac{\pi/4}{2\pi} = \frac{3}{8}$.

Given that the particle is above 0 at time 1, it is above 0 at time 2 with probability $\frac{3/8}{1/2} = \frac{3}{4}$ rather than $\frac{1}{2}$; the values B_1 and B_2 are dependent although the increments are independent.

The Reflection Principle for Brownian Motion

Theorem (reflection principle)

For $a > 0$ and $t > 0$: $\mathbb{P}(\max_{s \leq t} B_s \geq a) = 2\mathbb{P}(B_t \geq a) = \mathbb{P}(|B_t| \geq a)$.

The maximum up to time t has the same distribution as $|B_t|$, a single column of the table.

- This is the limit of slide 8; the term $\mathbb{P}(S_n = m)$ vanishes under the $\sqrt{n}$ scaling. Example:
 $\mathbb{P}(\max_{s \leq 1} B_s \geq 1) = 2(1 - \Phi(1)) \approx 0.317$; a direct computation would involve a continuum of columns.

Corollary (maximum and first hitting time)

Let $M_t = \max_{s \leq t} B_s$ and $T_a: \Omega \rightarrow [0, \infty]$, $T_a = \min\{s \geq 0 : B_s = a\}$. Then M_t has the distribution of $|B_t|$, and

$$\mathbb{P}(T_a \leq t) = \mathbb{P}(M_t \geq a) = 2(1 - \Phi(a/\sqrt{t})) \xrightarrow{t \rightarrow \infty} 1, \quad \text{yet} \quad \mathbb{E}[T_a] = \int_0^\infty \mathbb{P}(T_a > t) dt = \infty,$$

since $\mathbb{P}(T_a > t) = 2\Phi(a/\sqrt{t}) - 1 \approx a\sqrt{2/(\pi t)}$ for large t .

Every level is reached with probability one, but the mean time to reach it is infinite, as for the fair walk on slide 35.

Why the Paths Are Nowhere Differentiable

Theorem (nowhere differentiable paths)

With probability one, $t \mapsto B_t(\omega)$ is nowhere differentiable (Paley, Wiener, Zygmund, 1933; not proved here).

No path of B has a derivative at any time.

Why

By property (iii) of the definition (slide 15), $\frac{B_{t+h}-B_t}{h} \sim N\left(0, \frac{1}{h}\right)$: the difference quotient has standard deviation $1/\sqrt{h} \rightarrow \infty$ as $h \rightarrow 0$, so it cannot converge.

The Squared Increments Sum to t

Theorem

For $t > 0$ and $\Delta_k = B_{kt/n} - B_{(k-1)t/n}$, $k = 1, \dots, n$: $\mathbb{E}\left[\sum_{k=1}^n \Delta_k^2\right] = t$ and

$$\text{Var}\left(\sum_{k=1}^n \Delta_k^2\right) = \frac{2t^2}{n} \rightarrow 0.$$

Proof

The Δ_k are independent $N(0, \frac{t}{n})$ with $\mathbb{E}[\Delta_k^2] = \frac{t}{n}$ and $\text{Var}(\Delta_k^2) = 2(\frac{t}{n})^2$; sum over k . □

The squared increments of a Brownian path over $[0, t]$ sum to about t ; for a differentiable function they tend to 0 as $n \rightarrow \infty$.

- The n steps of S_n have squared increments $(\pm 1)^2$ summing to n , so the rescaled walk $B^{(n)}$ of slide 13 has $\sum (\Delta B^{(n)})^2 = n \cdot \frac{1}{n} = 1$ over $[0, 1]$, for every n , and so does the limit.
- The increments themselves do not add up: $\mathbb{E}|\Delta_k| = \sqrt{2t/(\pi n)}$ (the mean of $|N(0, \sigma^2)|$ is $\sigma\sqrt{2/\pi}$), so $\mathbb{E}\left[\sum_{k=1}^n |\Delta_k|\right] = \sqrt{2tn/\pi} \rightarrow \infty$: over any interval the path has infinite length.

Risk of Ruin: The Problem

A gambler with fortune $\mathcal{L}i$ bets $\mathcal{L}1$ per round, winning $\mathcal{L}1$ with probability p and losing it with probability $q = 1 - p$, rounds independent. She stops when her fortune reaches 0 (**ruin**) or a target $N > i$.

The opening question

Red at American roulette: $p = \frac{18}{38}$. She starts with $\mathcal{L}50$, aims for $\mathcal{L}100$, and stakes one pound per spin. We want the probability that she reaches the target, and the expected number of spins the game lasts.

Direct computation

The probability of reaching N is $\sum_n \mathbb{P}(\text{the game ends at round } n \text{ with fortune } N)$, a sum over paths of every length n that stay strictly between 0 and N until the last step. The summands have no closed form and there are infinitely many of them; the event is not determined by the fortune at any fixed time.

The question concerns where the path leaves an interval rather than its value at a fixed time; the same question arises for an insurer's reserves and for the length of a queue.

Risk of Ruin: The Setting

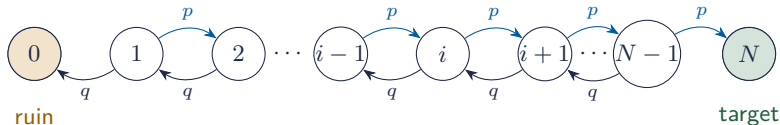
Definition

With $\xi_1, \xi_2, \dots$ the Bernoulli process of slide 6, the fortune is the walk started at i and stopped on exit from $\{1, \dots, N-1\}$:

$$X_n = i + S_n \text{ for } n \leq T, \quad T = \min\{n \geq 0 : X_n \in \{0, N\}\}, \quad h_i = \mathbb{P}_i(X_T = N).$$

The function $h: \{0, 1, \dots, N\} \rightarrow [0, 1]$ is the **probability of reaching the target** from fortune i ; $1 - h_i$ is the **probability of ruin**.

The fortune is a random walk on $\{0, \dots, N\}$ stopped at the endpoints, and h_i is the probability that the walk started at i reaches N before 0.



- T is a random variable $\Omega \rightarrow \{0, 1, 2, \dots\} \cup \{\infty\}$; that $T < \infty$ with probability one follows from the analysis below (the expected duration is finite).

First-Step Analysis: The Ruin Equation

Theorem

The function h satisfies

$$h_i = p h_{i+1} + q h_{i-1} \quad (0 < i < N), \quad h_0 = 0, \quad h_N = 1.$$

After one round the fortune is $i + 1$ or $i - 1$ and the game continues as if started from there, so h_i is a weighted average of h_{i+1} and h_{i-1} .

Proof

$h_0 = 0$ and $h_N = 1$ because in these states the game has already ended. For $0 < i < N$, condition on the first round (law of total probability with the two rows $\{\xi_1 = +1\}$ and $\{\xi_1 = -1\}$):

$$h_i = \mathbb{P}_i(X_T = N \mid \xi_1 = +1) p + \mathbb{P}_i(X_T = N \mid \xi_1 = -1) q.$$

On the row $\{\xi_1 = +1\}$ the fortune is $i + 1$ after round one and, by independent increments, the remaining rounds $\xi_2, \xi_3, \dots$ form a Bernoulli process independent of ξ_1 : the conditional probability of reaching N is h_{i+1} . Likewise h_{i-1} on the other row. □

- This is the argument of the Conditional Expectation lecture (Application 2) and of slide 11, applied to a probability; it gives $N - 1$ linear equations in $h_1, \dots, h_{N-1}$.

The Fair Game: $h_i = i/N$

Theorem ($p = q = \frac{1}{2}$)

$$h_i = \frac{i}{N} \text{ for } i = 0, \dots, N. \quad \text{Probability of ruin: } 1 - \frac{i}{N}.$$

In a fair game the probability of reaching N from i equals the fraction i/N of the target already held.

Proof

The equation $h_i = \frac{1}{2}h_{i+1} + \frac{1}{2}h_{i-1}$ says $h_{i+1} - h_i = h_i - h_{i-1}$: consecutive differences are equal, so h is linear in i , $h_i = A + Bi$. Then $h_0 = 0$ gives $A = 0$ and $h_N = 1$ gives $B = \frac{1}{N}$. Uniqueness: if h, h' both solve the system, $d = h - h'$ is linear with $d_0 = d_N = 0$, hence $d = 0$. $\square$

Second proof: the fair game preserves expected fortune

Each round has mean gain 0, so $\mathbb{E}[X_T] = X_0 = i$ (this needs an argument for the random time T , which we omit). But $X_T \in \{0, N\}$, so $\mathbb{E}[X_T] = N h_i$. Hence $h_i = i/N$.

- £50 to £100 at a fair game: $\frac{1}{2}$. £3 to £10: $\frac{3}{10}$. £1 to £10: $\frac{1}{10}$, and ruin $\frac{9}{10}$.

The Unfair Game: $h_i = \frac{r^i - 1}{r^N - 1}$ with $r = q/p$

Theorem ($p \neq q$)

With $r = q/p \neq 1$: $h_i = \frac{r^i - 1}{r^N - 1}$, probability of ruin $1 - h_i = \frac{r^N - r^i}{r^N - 1}$.

The probabilities now depend on i through the geometric sequence r^i rather than linearly, so a small disadvantage per round accumulates over the many rounds needed to reach the target.

Proof

Try $h_i = r^i$ in $h_i = p h_{i+1} + q h_{i-1}$: the right side is $r^{i-1}(pr^2 + q)$, and $pr^2 + q = \frac{q^2}{p} + q = \frac{q(q+p)}{p} = r$, so it equals r^i . Constants also solve the equation. Hence $h_i = A + Br^i$ solves it for every A, B , and since a solution is fixed by h_0, h_1 (the equation determines h_{i+1} from h_i, h_{i-1}), these are all the solutions. Now $h_0 = 0$ gives $A = -B$ and $h_N = 1$ gives $B(r^N - 1) = 1$. $\square$

$$p = 0.4, r = 1.5$$

From £5 to £10: $h_5 = \frac{1.5^5 - 1}{1.5^{10} - 1} = \frac{32}{275} \approx 0.116$, against $\frac{1}{2}$ in the fair game. From £3 to £10: $h_3 \approx 0.042$, against 0.3.

Expected Duration of the Game

Let $D_i = \mathbb{E}_i[T]$, so $D: \{0, \dots, N\} \rightarrow [0, \infty)$.

Theorem

$D_i = 1 + p D_{i+1} + q D_{i-1}$ for $0 < i < N$, $D_0 = D_N = 0$, and the solution is

$$D_i = i(N-i) \quad (p = q), \quad D_i = \frac{i}{q-p} - \frac{N}{q-p} \cdot \frac{1-r^i}{1-r^N} \quad (p \neq q, r = q/p).$$

In the fair game started from the midpoint the expected duration is $N^2/4$ rounds, the square of the distance to the boundary, as on slide 12.

Proof

The equation is first-step analysis with one round paid before moving to $i \pm 1$, as on slide 11. For $p = q$ it is the equation of slide 12 on $\{0, \dots, N\}$; $D_i = i(N-i)$ satisfies it since $\frac{1}{2}((i+1)(N-i-1) + (i-1)(N-i+1)) = i(N-i) - 1$, and it vanishes at 0 and N . For $p \neq q$: $D_i = \frac{i}{q-p}$ is one solution of the equation with the $+1$ ($\frac{p(i+1)+q(i-1)}{q-p} = \frac{i-(q-p)}{q-p} = \frac{i}{q-p} - 1$); add the general solution $A + Br^i$ of the homogeneous equation and fit $D_0 = D_N = 0$. Uniqueness as before. $\square$

- Fair, £50 to £100: 2500 rounds on average. Roulette ($p = \frac{18}{38}$), same stakes: about 940 rounds. The unfavourable game is both more likely to end in ruin and shorter on average.

The Casino's Edge in Numbers

$i \rightarrow N$ (in units of one stake)	p	$\mathbb{P}(\text{target})$	$\mathbb{E}[T]$
$50 \rightarrow 100$	0.5	0.500	2500
$50 \rightarrow 100$	0.49	0.119	1904
$50 \rightarrow 100$	18/38	0.005	940
$10 \rightarrow 20$	18/38	0.259	92
$5 \rightarrow 10$	18/38	0.371	24
$1 \rightarrow 2$	18/38	0.474	1

At a 1% disadvantage, doubling £50 one pound at a time succeeds with probability about 0.12; the last three rows are £50 → £100 with stakes of £5, £10 and £50, and with fewer rounds the disadvantage accumulates less.

Why Doubling Fails

Doubling (the “martingale” system)

Bet 1; after each loss double the stake; after a win collect the net gain of £1 and restart. With capital $2^k - 1$ the gambler can absorb $k - 1$ consecutive losses; k losses in a row exhaust the capital. Per cycle: gain +1 with probability $1 - q^k$, lose $2^k - 1$ with probability q^k :

$$\mathbb{E}[\text{gain}] = (1 - q^k) - q^k(2^k - 1) = 1 - (2q)^k.$$

This is 0 if $q = \frac{1}{2}$ and negative if $q > \frac{1}{2}$. The strategy changes the distribution of the outcome, to many small wins and a rare large loss, but not its mean.

- Roulette, $k = 7$ (capital £127): win £1 with probability 0.989, lose £127 with probability 0.011; mean -£0.43 per cycle.

Doubling produces many small wins and a rare large loss; the mean gain per cycle is negative whenever each bet has negative mean.

The Infinite Target: An Opponent With Unlimited Capital

Let $N \rightarrow \infty$ with i fixed: the gambler plays on until ruin or forever.

Theorem

Let $\rho_i = \mathbb{P}_i(\text{ruin, ever}) = \lim_{N \rightarrow \infty} (1 - h_i)$. Then

$$\rho_i = 1 \quad (p \leq q), \quad \rho_i = \left(\frac{q}{p}\right)^i \quad (p > q).$$

In the fair case, moreover, $\mathbb{E}[\text{time to ruin}] = \lim_N i(N - i) = \infty$.

Against an opponent with unlimited capital, a fair or unfavourable game ends in ruin with probability one; in a favourable game the probability of ruin is $(q/p)^i$, which decreases geometrically in the initial fortune i .

Proof

Fair: $1 - h_i = 1 - \frac{i}{N} \rightarrow 1$. Unfair with $r = q/p$: $1 - h_i = \frac{r^N - r^i}{r^N - 1}$. If $r > 1$ ($p < q$), divide by r^N :
 $\rightarrow \frac{1 - 0}{1 - 0} = 1$. If $r < 1$ ($p > q$), $r^N \rightarrow 0$ and $1 - h_i \rightarrow \frac{0 - r^i}{0 - 1} = r^i$. □

The Infinite Target: Three Regimes

game	$\mathbb{P}(\text{ruin})$ from i	$\mathbb{E}[\text{time to ruin}]$	the walk $i + S_n$
unfavourable, $p < q$	1	$\frac{i}{q-p} < \infty$	$\mathbb{E}[\xi_n] < 0$: reaches 0 with probability 1, in finite mean time
fair, $p = q$	1	∞	$\mathbb{E}[\xi_n] = 0$: reaches 0 with probability 1, but with infinite mean time
favourable, $p > q$	$(q/p)^i < 1$	∞	$\mathbb{E}[\xi_n] > 0$: reaches 0 with probability $(q/p)^i < 1$

- $p = 0.6$, £5 in hand: ruin with probability $(\frac{2}{3})^5 \approx 0.13$; otherwise the fortune drifts to infinity.
- Roulette, £50 in hand, no target: ruin certain, after $\frac{50}{2/38} = 950$ spins on average.
- Fair game: ruin is certain, but the expected time to ruin is infinite. This is the statement that the fair walk reaches any level with probability one but with infinite expected time; it reappears as *null recurrence* on slide 62.
- ρ_i for $p > q$ is the probability that a walk with upward drift ever falls by i below its start: the **hitting probability** of level $-i$. Hitting probabilities are the subject of slides 63–66, where the ruin problem returns as a special case.

The three regimes differ in whether ruin is certain and, if it is, whether the expected time to ruin is finite.

Return to the Opening Problem

Red at roulette, $p = \frac{18}{38}$, $r = \frac{20}{18} = \frac{10}{9}$. From £50, one pound per spin:

target N	$h_{50} = \frac{r^{50} - 1}{r^N - 1}$	ruin	$\mathbb{E}[T]$ (spins)
100 (double)	0.005	0.995	940
60 (win £10)	0.348	0.652	554
55 (win £5)	0.589	0.411	334
51 (win £1)	0.900	0.100	78

- Doubling £50 at £1 a spin succeeds with probability about $1/195$, and the game lasts about 940 spins on average. Staking all £50 on a single spin succeeds with probability $\frac{18}{38} \approx 0.474$.
- The plan “win £1 and stop” fails with probability 0.1005, and each failure costs £50; the mean gain is $0.8995 \cdot 1 - 0.1005 \cdot 50 \approx -£4.12$, which equals $(p - q) \mathbb{E}[T] = -\frac{2}{38} \cdot 78.4$. No staking plan has positive mean gain, since every plan is a sum of bets each of mean $-\frac{2}{38}$ per pound staked.

The entries are the values of h_{50} and D_{50} given by the formulas of slides 30 and 31 for each target N .

Markov Chains: The Problem

A town's weather is Sunny, Cloudy or Rainy each day, and tomorrow's weather depends on today's according to the table below. It is sunny today.

1. What is the probability that it is rainy a week from now?
2. In the long run, what fraction of days are sunny?
3. How many days until the next rainy day, on average?

Direct computation

Question 1 is a sum over the $3^6 = 729$ possible weather sequences for the six intermediate days. Question 2 involves a limit over sequences of every length. Question 3 concerns a waiting time whose distribution has no simple closed form.

In this model tomorrow's weather depends on the past only through today's; this property reduces all three questions to computations with the 3×3 matrix.

today \ tomorrow	S	C	R
S	0.7	0.2	0.1
C	0.3	0.4	0.3
R	0.2	0.4	0.4

Row S reads: after a sunny day, tomorrow is sunny with probability 0.7, cloudy 0.2, rainy 0.1. Each row sums to 1.

Markov Chain: Definition

Definition

Let S be a finite or countable set (the **state space**). A process $(X_n)_{n \geq 0}$ with $X_n: \Omega \rightarrow S$ is a **Markov chain** if for all $n \geq 0$ and all states $i_0, \dots, i_{n-1}, i, j \in S$ with $\mathbb{P}(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i) > 0$,

$$\mathbb{P}(X_{n+1} = j \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i) = \mathbb{P}(X_{n+1} = j \mid X_n = i) \quad (\text{Markov property}),$$

and it is **homogeneous** if the right side does not depend on n . Then the **transition probabilities** are the function

$$p: S \times S \rightarrow [0, 1], \quad p(i, j) = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

The conditional distribution of the next state given the entire past depends only on the present state, and in a homogeneous chain it is the same at every time.

- The two arguments of p play different roles: i is the *current* state (the conditioning event $\{X_n = i\}$, the row); j is the *next* state (the column). $p(i, j)$ is the probability of the step $i \rightarrow j$.
- Random walk: $S = \mathbb{Z}$, $p(i, i+1) = p$, $p(i, i-1) = q$, all other $p(i, j) = 0$. The Markov property holds because $S_{n+1} = S_n + \xi_{n+1}$ with ξ_{n+1} independent of the past. Gambler's ruin: the same on $S = \{0, \dots, N\}$ with $p(0, 0) = p(N, N) = 1$.
- All chains below are homogeneous.

The Transition Matrix: A Table Whose Rows Are Conditional Distributions

Definition

The **transition matrix** is p written as a table,
 $\mathbf{P} = (p(i, j))_{i, j \in S}$: row i , column j . It is a **stochastic matrix**:

$$p(i, j) \geq 0 \quad \text{and} \quad \sum_{j \in S} p(i, j) = 1 \quad \text{for every } i \in S.$$

The **initial distribution** is the row vector $\lambda: S \rightarrow [0, 1]$,
 $\lambda_i = \mathbb{P}(X_0 = i)$, $\sum_i \lambda_i = 1$.

Row i of $\mathbf{P}$ is the conditional pmf of X_{n+1} given $X_n = i$, as a row of the two-dice table is.

- Row i , as a function $j \mapsto p(i, j)$, is $p_{X_{n+1}|X_n}(\cdot | i)$: non-negative, sums to 1. Column sums are unconstrained.
- $\mathbb{E}[f(X_{n+1}) | X_n = i] = \sum_j p(i, j)f(j)$: the row average of f (compare $\mathbb{E}[X | Y = y]$, lecture 5).

		next state j		
		S	C	R
current state i	S	0.7	0.2	0.1
	C	0.3	0.4	0.3
	R	0.2	0.4	0.4

The weather chain. The shaded row is $p_{X_{n+1}|X_n}(\cdot | C)$: the distribution of tomorrow given a cloudy today.

Paths Have Product Probabilities

Theorem

For a Markov chain with initial distribution λ and transition matrix $\mathbf{P}$, and any states $i_0, i_1, \dots, i_n$,

$$\mathbb{P}(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) = \lambda_{i_0} p(i_0, i_1) p(i_1, i_2) \cdots p(i_{n-1}, i_n).$$

Conversely, a process whose joint pmfs have this form is a Markov chain with transition matrix $\mathbf{P}$.

The probability of a row of the table of paths is the start probability times the product of the step probabilities along the row; λ and $\mathbf{P}$ together determine every finite-dimensional distribution.

Proof

Multiplication rule:

$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = \mathbb{P}(X_0 = i_0) \prod_{k=1}^n \mathbb{P}(X_k = i_k \mid X_0 = i_0, \dots, X_{k-1} = i_{k-1})$, and each factor is $p(i_{k-1}, i_k)$ by the Markov property. For the converse, divide two such products to get the conditional probability. □

Weather chain (slide 39), sunny today

$$\mathbb{P}(X_1 = \text{C}, X_2 = \text{R} \mid X_0 = \text{S}) = p(\text{S}, \text{C}) p(\text{C}, \text{R}) = 0.2 \cdot 0.3 = 0.06;$$

$$\mathbb{P}(X_1 = \text{R}, X_2 = \text{R}, X_3 = \text{S} \mid X_0 = \text{S}) = 0.1 \cdot 0.4 \cdot 0.2 = 0.008.$$

What the Markov Property Is Not

- **It is not independence.** X_{n+1} may depend strongly on X_n ; the property says that the earlier states $X_0, \dots, X_{n-1}$ carry no further information once X_n is known.
- **It does not say the past is irrelevant.** In general $\mathbb{P}(X_{n+1} = j \mid X_{n-1} = k) \neq \mathbb{P}(X_{n+1} = j)$; the past is irrelevant only conditionally on the present.
- **Not every process has it.** Toss a coin and let X_n be the number of heads among tosses $n-1$ and n . Given $X_n = 1$, the distribution of X_{n+1} depends on which of the two tosses was the head: $\mathbb{P}(X_{n+1} = 2 \mid X_n = 1, X_{n-1} = 0) = \frac{1}{2}$ but $\mathbb{P}(X_{n+1} = 2 \mid X_n = 1, X_{n-1} = 2) = 0$. It is not Markov; the process recording the last two tosses is.
- **It depends on what is recorded as the state.** A process that is not Markov often becomes Markov when the state is enlarged to include the relevant part of the past (the last two tosses; position and velocity; today's weather and yesterday's).

Test for the Markov property

Check that $\mathbb{P}(X_{n+1} = j \mid X_n = i, \text{ anything about } X_0, \dots, X_{n-1})$ does not change with the “anything”. In models of the form $X_{n+1} = F(X_n, \xi_{n+1})$ with ξ_{n+1} independent of $X_0, \dots, X_n$ it holds automatically; the random walk, the Ehrenfest urn and the walks on graphs below are all of this form.

The Markov property states that, for predicting the future, the present state is a sufficient summary of the past.

A Gallery of Chains (1)

Vowels and consonants

Model a text as a sequence of V and C. After a vowel, the next letter is a vowel with probability 0.2; after a consonant, with probability 0.5.

$$S = \{V, C\}, \quad \mathbf{P} = \begin{pmatrix} 0.2 & 0.8 \\ 0.5 & 0.5 \end{pmatrix}.$$

This two-state chain is used in the Entropy lecture for the entropy-rate example.

Gambler's ruin

$S = \{0, \dots, N\}$, $p(i, i+1) = p$, $p(i, i-1) = q$ for $0 < i < N$, and $p(0,0) = p(N,N) = 1$: the diagram of slide 27. States 0 and N are **absorbing**: once the chain enters one of them it remains there.

The Ehrenfest urn (gas between two chambers)

N numbered balls in two urns. Each step, pick a ball uniformly at random and move it to the other urn. X_n is the number of balls in urn 1: $S = \{0, \dots, N\}$,

$$p(i, i-1) = \frac{i}{N}, \quad p(i, i+1) = \frac{N-i}{N}.$$

The more balls an urn holds, the more likely the next move is out of it.

$$N = 4: \quad \mathbf{P} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ \frac{1}{4} & 0 & \frac{3}{4} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{3}{4} & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

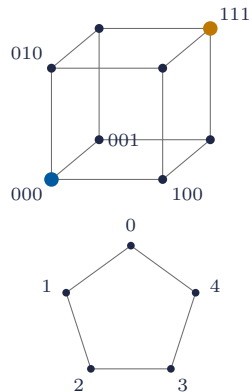
A Gallery of Chains (2): Walks on Graphs

Random walk on a graph

Vertices S ; from vertex i move to one of its $d(i)$ neighbours uniformly: $p(i, j) = 1/d(i)$ if j is a neighbour of i , else 0.

- **Cycle** of length L : $S = \{0, \dots, L-1\}$, $p(i, i \pm 1 \bmod L) = \frac{1}{2}$. This is the random walk on $\{0, \dots, L-1\}$ with the two ends joined.
- **Cube**: $S = \{0, 1\}^3$, the 8 corners; neighbours differ in one coordinate, so $p(i, j) = \frac{1}{3}$ for each of the 3 neighbours. A move flips one coordinate.
- **The web**: pages as vertices, links as directed edges, $p(i, j) = 1/(\text{number of links out of } i)$ for each page j that i links to. This is the **random surfer** of slides 54–55.

On a graph, the transition matrix is the adjacency table with each row rescaled to sum to 1.



Left: the cube; from 000, the three neighbours 100, 010, 001. Right: the cycle $L = 5$.

n -Step Transition Probabilities

As on slide 11, $\mathbb{P}_i(\cdot) = \mathbb{P}(\cdot \mid X_0 = i)$ and $\mathbb{E}_i[\cdot] = \mathbb{E}[\cdot \mid X_0 = i]$ refer to the chain started at i (the case $\lambda = \text{point mass at } i$).

Definition

The **n -step transition probabilities** are the function $p^{(n)}: S \times S \rightarrow [0, 1]$,

$$p^{(n)}(i, j) = \mathbb{P}_i(X_n = j) = \mathbb{P}(X_{m+n} = j \mid X_m = i) \quad (\text{any } m, \text{ by homogeneity}),$$

with $p^{(1)} = p$ and $p^{(0)}(i, j) = \mathbf{1}\{i = j\}$. As before, i is the current state (row) and j the state n steps later (column).

The n -step probabilities form a table in which row i is the conditional distribution of X_{m+n} given $X_m = i$.

Weather: $p^{(2)}(S, R)$ by hand

Sum over tomorrow's weather k :

$p^{(2)}(S, R) = \sum_k p(S, k) p(k, R) = 0.7 \cdot 0.1 + 0.2 \cdot 0.3 + 0.1 \cdot 0.4 = 0.17$. The three terms are the sequences SSR, SCR, SRR, each a product as on slide 40.

- This sum is the (S, R) entry of $\mathbf{PP}$; the next slide shows that this holds in general.

Chapman–Kolmogorov: n Steps Is the n -th Power

Theorem (Chapman–Kolmogorov)

For all $m, n \geq 0$ and $i, j \in S$: $p^{(m+n)}(i, j) = \sum_{k \in S} p^{(m)}(i, k) p^{(n)}(k, j)$. Equivalently, writing

$\mathbf{P}^{(n)} = (p^{(n)}(i, j))_{i, j}$ for the table of n -step probabilities: $\mathbf{P}^{(n)} = \mathbf{P}^n$, the n -th matrix power.

A path from i to j in $m + n$ steps is in some state k at time m ; summing over k gives the formula, which has the form of a matrix product.

Proof

Total probability with the rows $\{X_m = k\}$, $k \in S$:

$$\mathbb{P}_i(X_{m+n} = j) = \sum_k \mathbb{P}_i(X_m = k) \mathbb{P}_i(X_{m+n} = j \mid X_m = k) = \sum_k p^{(m)}(i, k) p^{(n)}(k, j),$$

the second factor by the Markov property and homogeneity (the past before time m , including $X_0 = i$, drops out). With $m = 1$ this is $\mathbf{P}^{(n+1)} = \mathbf{P} \mathbf{P}^{(n)}$, so by induction $\mathbf{P}^{(n)} = \mathbf{P}^n$. $\square$

- The case $m = 1$ is first-step analysis and $n = 1$ a last-step decomposition; numerically, $\mathbf{P}^n$ is computed by repeated squaring. Rows of $\mathbf{P}^n$ are conditional distributions, so $\mathbf{P}^n$ is stochastic.

The Distribution at Time n : Row Vector Times $\mathbf{P}^n$

Theorem

If X_0 has distribution λ (a row vector indexed by S), then X_n has distribution $\lambda \mathbf{P}^n$:

$$\mathbb{P}(X_n = j) = \sum_{i \in S} \lambda_i p^{(n)}(i, j) = (\lambda \mathbf{P}^n)_j.$$

The distribution at time n is the initial distribution multiplied by $\mathbf{P}$ once for each step; each multiplication forms a weighted average of the rows of $\mathbf{P}$ with the current probabilities as weights.

Proof

Total probability over the rows $\{X_0 = i\}$: $\mathbb{P}(X_n = j) = \sum_i \mathbb{P}(X_0 = i) \mathbb{P}(X_n = j \mid X_0 = i)$. □

- Every finite-dimensional distribution follows: $\mathbb{P}(X_m = i, X_{m+n} = j) = (\lambda \mathbf{P}^m)_i p^{(n)}(i, j)$, and so on along any sequence of times.
- $\lambda \mathbf{P}^n$ is a row vector, a distribution on S , while $\mathbf{P}^n$ is a matrix, a table of conditional distributions. Multiplication by $\mathbf{P}$ advances a distribution by one step in time.

Return to Question 1: Rainy a Week From a Sunny Day

$\lambda = (1, 0, 0)$, so the distribution of X_7 is row S of $\mathbf{P}^7$.

$$\mathbf{P}^2 = \begin{pmatrix} 0.57 & 0.26 & 0.17 \\ 0.39 & 0.34 & 0.27 \\ 0.34 & 0.36 & 0.30 \end{pmatrix}, \quad \mathbf{P}^3 = \begin{pmatrix} 0.511 & 0.286 & 0.203 \\ 0.429 & 0.322 & 0.249 \\ 0.406 & 0.332 & 0.262 \end{pmatrix}, \quad \mathbf{P}^7 = \begin{pmatrix} 0.4637 & 0.3068 & 0.2296 \\ 0.4601 & 0.3083 & 0.2316 \\ 0.4591 & 0.3087 & 0.2321 \end{pmatrix}.$$

Answer

$\mathbb{P}_S(X_7 = R) = p^{(7)}(S, R) \approx 0.230$. The sum over 729 six-day sequences has been organised into six matrix multiplications.

- The three rows of $\mathbf{P}^7$ are almost equal: after a week the distribution of the weather depends very little on today's weather. The common limiting row, $(0.4615, 0.3077, 0.2308)$ to four places from $\mathbf{P}^{20}$, is the subject of the next section.
- Comparing $\mathbf{P}^2$ with $\mathbf{P}$: after a rainy day the probability of sun is 0.2 for the next day and 0.34 for the day after; the rows of $\mathbf{P}^n$ approach one another as n grows.

The distribution of the chain at any future time is obtained from the powers of $\mathbf{P}$.

Powers of a 2×2 Chain: The Rows Converge

Vowel/consonant chain

$$\mathbf{P} = \begin{pmatrix} 0.2 & 0.8 \\ 0.5 & 0.5 \end{pmatrix}, \quad \mathbf{P}^2 = \begin{pmatrix} 0.44 & 0.56 \\ 0.35 & 0.65 \end{pmatrix}, \quad \mathbf{P}^3 = \begin{pmatrix} 0.368 & 0.632 \\ 0.395 & 0.605 \end{pmatrix}, \quad \mathbf{P}^{10} = \begin{pmatrix} 0.3846 & 0.6154 \\ 0.3846 & 0.6154 \end{pmatrix}.$$

Both rows approach $(\frac{5}{13}, \frac{8}{13}) = (0.384615\dots, 0.615385\dots)$; after ten letters the probability of a vowel no longer depends appreciably on whether the first letter was a vowel or a consonant.

Exact formula

$\mathbf{P}$ has eigenvalues 1 and $0.2 + 0.5 - 1 = -0.3$ (trace minus 1), and

$$p^{(n)}(\mathbf{V}, \mathbf{V}) = \frac{5}{13} + \frac{8}{13}(-0.3)^n, \quad p^{(n)}(\mathbf{C}, \mathbf{V}) = \frac{5}{13} - \frac{5}{13}(-0.3)^n.$$

For $n = 1$ these give $\frac{5}{13} - \frac{2.4}{13} = 0.2$ and $\frac{5}{13} + \frac{1.5}{13} = 0.5$, as required. The difference from the limit is multiplied by -0.3 at each step.

The remaining sections identify the limit row, give conditions under which it exists, and estimate the speed of convergence.

Stationary Distribution: Definition

Definition

A **stationary distribution** of a chain with transition matrix $\mathbf{P}$ on S is a function $\pi: S \rightarrow [0, 1]$ with $\sum_{i \in S} \pi_i = 1$ and

$$\pi \mathbf{P} = \pi, \quad \text{i.e.} \quad \sum_{i \in S} \pi_i p(i, j) = \pi_j \quad \text{for every } j \in S.$$

A stationary distribution is one that the chain preserves: if X_0 has distribution π then every X_n has distribution π .

Theorem

If $X_0 \sim \pi$ and π is stationary, then $X_n \sim \pi$ for every n , and (X_n, X_{n+1}) has the same joint distribution for every n .

Proof

$\pi \mathbf{P}^n = (\pi \mathbf{P}) \mathbf{P}^{n-1} = \pi \mathbf{P}^{n-1} = \dots = \pi$, then the previous slide. $\mathbb{P}(X_n = i, X_{n+1} = j) = \pi_i p(i, j)$ does not involve n . □

Stationary Distribution: Two Readings and Existence

Balance

$\sum_i \pi_i p(i, j)$ is the probability mass flowing into j in one step from the distribution π ; stationarity says it equals the mass π_j already there, for every j . Mass leaves j at rate $\pi_j \sum_k p(j, k) = \pi_j$, so inflow equals outflow at every state.

Left eigenvector

$\pi \mathbf{P} = \pi$ says π is a left eigenvector of $\mathbf{P}$ for the eigenvalue 1, normalised to sum to 1. Every stochastic matrix has the eigenvalue 1, because the column vector $\mathbf{1}$ of ones satisfies $\mathbf{P}\mathbf{1} = \mathbf{1}$ (row sums), and left and right eigenvalues coincide.

Theorem (existence)

Every Markov chain on a finite state space has at least one stationary distribution.

A left eigenvector for the eigenvalue 1 exists by the argument above; that it can be chosen with non-negative entries is the content of the theorem.

- The theorem is not proved here (Perron–Frobenius, or a fixed-point argument). Uniqueness can fail, as for the gambler's ruin chain, which has infinitely many. On an infinite S existence can fail: the random walk on $\mathbb{Z}$ has none, since any solution of $\pi \mathbf{P} = \pi$ is constant and cannot sum to 1.

Computing π : A Linear System

$\pi \mathbf{P} = \pi$ is $|S|$ linear equations in $|S|$ unknowns, but they are dependent (they sum to the identity $\sum_j \pi_j = \sum_i \pi_i$); drop one and add $\sum_i \pi_i = 1$.

Weather, question 2: the long-run fraction of sunny days

Unknowns π_S, π_C, π_R . Columns S and C of $\pi \mathbf{P} = \pi$, and normalisation:

$$0.7\pi_S + 0.3\pi_C + 0.2\pi_R = \pi_S, \quad 0.2\pi_S + 0.4\pi_C + 0.4\pi_R = \pi_C, \quad \pi_S + \pi_C + \pi_R = 1.$$

The first gives $3\pi_S = 3\pi_C + 2\pi_R$, the second $6\pi_C = 2\pi_S + 4\pi_R$; solving,

$$\pi = \left(\frac{6}{13}, \frac{4}{13}, \frac{3}{13} \right) \approx (0.4615, 0.3077, 0.2308),$$

which agrees with the common row of $\mathbf{P}^{20}$ on slide 47. About 46% of days are sunny in the long run (the precise sense of “long run” is given on slide 74).

Vowel/consonant

$0.2\pi_V + 0.5\pi_C = \pi_V$, i.e. $0.8\pi_V = 0.5\pi_C$, with $\pi_V + \pi_C = 1$: $\pi = \left(\frac{5}{13}, \frac{8}{13} \right)$, the limit row of slide 48 and the weights in the Entropy lecture's entropy rate.

Detailed Balance and Reversible Chains

Theorem

If $\pi: S \rightarrow [0, 1]$ sums to 1 and satisfies the **detailed balance** equations

$$\pi_i p(i, j) = \pi_j p(j, i) \quad \text{for all } i, j \in S,$$

then π is stationary. Such a chain is called **reversible** (with respect to π).

If the flow along each edge balances in both directions, then the total flow into each state equals the flow out of it.

Proof

$$\sum_i \pi_i p(i, j) = \sum_i \pi_j p(j, i) = \pi_j \sum_i p(j, i) = \pi_j. \quad \square$$

- Stronger than stationarity, with $|S|^2$ equations, but each involves only two unknowns, so the system is solved edge by edge; if it has no solution, π must be found from $\pi \mathbf{P} = \pi$.
- Random walk on a graph: $\pi_i = d(i) / \sum_k d(k)$ satisfies $\pi_i p(i, j) = \frac{1}{\sum_k d(k)} = \pi_j p(j, i)$ on every edge. Cube and cycle: uniform.
- Started from π , a reversible chain has the same joint distributions when run backwards, since $\mathbb{P}(X_n = i, X_{n+1} = j) = \pi_i p(i, j) = \pi_j p(j, i) = \mathbb{P}(X_n = j, X_{n+1} = i)$.

Detailed Balance: The Ehrenfest Urn

The Ehrenfest urn: π is binomial

Transitions occur only between neighbouring states, so detailed balance is one equation for each edge: $\pi_i \frac{N-i}{N} = \pi_{i+1} \frac{i+1}{N}$, i.e. $\pi_{i+1} = \pi_i \frac{N-i}{i+1}$. Iterating from π_0 gives $\pi_i = \pi_0 \binom{N}{i}$, and $\sum_i \pi_i = 1$ forces $\pi_0 = 2^{-N}$:

$$\pi_i = \binom{N}{i} 2^{-N} \quad (\text{each ball in urn 1 independently with probability } \tfrac{1}{2}).$$

$N = 4$: $\pi = \frac{1}{16}(1, 4, 6, 4, 1)$. Check one edge: $\pi_1 p(1, 2) = \frac{4}{16} \cdot \frac{3}{4} = \frac{3}{16} = \frac{6}{16} \cdot \frac{1}{2} = \pi_2 p(2, 1)$.

Under the stationary distribution each ball is in urn 1 independently with probability $\frac{1}{2}$.

- Solving $\pi \mathbf{P} = \pi$ directly means solving a 5×5 system, or a 101×101 system for $N = 100$; detailed balance gives π by a recursion along the edges.
- $\pi_0 = 2^{-N}$, so by slide 61 the mean time between two moments when urn 1 is empty is 2^N steps: 16 for $N = 4$, about $1.3 \cdot 10^{30}$ for $N = 100$; meanwhile π is concentrated within a few multiples of $\sqrt{N}$ of $N/2$.

PageRank: The Stationary Distribution of a Random Surfer

Definition

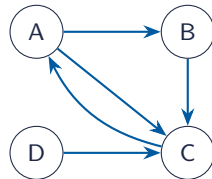
n web pages, links as directed edges. A **random surfer** moves as follows: with probability d (the damping factor, $d = 0.85$) follow a uniformly chosen link out of the current page; with probability $1 - d$ jump to a uniformly random page. Its transition matrix is

$$\mathbf{P} = d\mathbf{H} + \frac{1-d}{n}\mathbf{J}, \quad \mathbf{H}_{ij} = \frac{\mathbf{1}\{i \rightarrow j\}}{\#\{\text{links out of } i\}}, \quad \mathbf{J}_{ij} = 1.$$

The **PageRank** of page j is π_j , where $\pi: S \rightarrow [0, 1]$ is the stationary distribution of $\mathbf{P}$.

A page ranks highly if highly ranked pages link to it; the equation $\pi_j = \sum_i \pi_i p(i, j)$ makes this precise.

- The jump is needed because from a page with no out-links, or a group of pages with no exits, a surfer following links alone can never leave, and the chain with matrix $\mathbf{H}$ alone may have no unique π or may fail to converge (slides 56–71). With $d < 1$ every entry of $\mathbf{P}$ is positive, so the results below apply.



Four pages. Links $A \rightarrow B, C$; $B \rightarrow C$; $C \rightarrow A$; $D \rightarrow C$. $\mathbf{P}$ adds a 0.0375 chance of jumping to each page from each page.

In $\mathbf{H}_{ij}$, i is the page the surfer is on and j the page linked to; row sums are 1 (for a page with no out-links put $\mathbf{H}_{ij} = 1/n$ for all j).

PageRank: The Four-Page Web

$$\mathbf{H} = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{P} = 0.85 \mathbf{H} + 0.0375 \mathbf{J} = \begin{pmatrix} 0.0375 & 0.4625 & 0.4625 & 0.0375 \\ 0.0375 & 0.0375 & 0.8875 & 0.0375 \\ 0.8875 & 0.0375 & 0.0375 & 0.0375 \\ 0.0375 & 0.0375 & 0.8875 & 0.0375 \end{pmatrix}.$$

Solving $\pi \mathbf{P} = \pi$, $\sum_j \pi_j = 1$

$$\pi \approx (0.373, 0.196, 0.394, 0.038) \quad \text{for (A, B, C, D),} \quad \text{exactly } \pi_D = \frac{3}{80} = 0.0375.$$

C ranks first; three pages link to it, and one of them, A, is itself highly ranked. A is second although it has a single in-link, because that link comes from the top page and is C's only link. D has no in-links and receives only the jump mass $\frac{1-d}{n}$.

- The definition of rank is circular, since A ranks highly because C does and C because A does. The stationary equation expresses this circularity as a fixed-point condition, and the powers $\mathbf{P}^n$ compute the fixed point by iteration ($\mathbf{P}^{30}$ has all rows equal to π to four places).
- Without damping ($d = 1$), $\pi = (0.4, 0.2, 0.4, 0)$ is stationary for $\mathbf{H}$: a page with no in-links has rank 0.

Communicating Classes and Irreducibility

Definition

State i **leads to** j , written $i \rightarrow j$, if $p^{(n)}(i, j) > 0$ for some $n \geq 0$. States i, j **communicate**, $i \leftrightarrow j$, if $i \rightarrow j$ and $j \rightarrow i$. This is an equivalence relation on S ; its equivalence classes are the **communicating classes**. A class C is **closed** if $i \in C$ and $i \rightarrow j$ imply $j \in C$. The chain is **irreducible** if S is a single class.

Two states are in the same class if the chain can get from each to the other; a closed class is one the chain cannot leave; irreducible means every state is reachable from every other.

- $i \rightarrow j$ iff there is a path $i = i_0, i_1, \dots, i_n = j$ with every $p(i_{k-1}, i_k) > 0$: a directed path in the graph whose edges are the positive entries of $\mathbf{P}$. The classes are the strongly connected components of that graph.
- Weather, vowel/consonant, Ehrenfest, cycle, cube, PageRank with $d < 1$: irreducible.
- Gambler's ruin: three classes, $\{0\}$ and $\{N\}$ closed (absorbing states), $\{1, \dots, N-1\}$ not closed. Four-page web without damping: $\{A, B, C\}$ closed, $\{D\}$ not closed.
- A state in a class that is not closed is left permanently with positive probability at each visit, so it is visited only finitely often (next slides).

Periodicity

Definition

The **period** of state i is $d(i) = \gcd\{n \geq 1 : p^{(n)}(i, i) > 0\}$ (with $d(i) = \infty$ if the set is empty). State i is **aperiodic** if $d(i) = 1$. A chain is aperiodic if every state is.

The period is the greatest common divisor of the lengths of all round trips from i ; period 2 means every return takes an even number of steps.

Theorem

All states in a communicating class have the same period. If $p(i, i) > 0$ for some i in a class, the class is aperiodic.

Proof

Let $i \leftrightarrow j$ with $p^{(a)}(i, j) > 0$, $p^{(b)}(j, i) > 0$. Every round trip from j of length n gives round trips from i of lengths $a + b$ and $a + n + b$, so $d(i)$ divides both, hence divides n ; thus $d(i) \leq d(j)$, and symmetrically. If $p(i, i) > 0$ then 1 is a round-trip length. □

Periodicity: The Random Walk on the Cube

Random walk on the cube

Each step flips one coordinate, so the parity of the number of 1s alternates: even at even times, odd at odd times. Hence $p^{(n)}(000, 000) = 0$ for odd n , $d = 2$, and

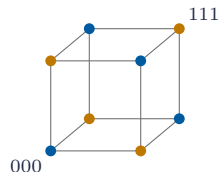
$$(\mathbf{P}^{10})_{000, \cdot} \approx \frac{1}{4}(1, 0, 0, 1, 0, 1, 1, 0),$$

$$(\mathbf{P}^{11})_{000, \cdot} \approx \frac{1}{4}(0, 1, 1, 0, 1, 0, 0, 1)$$

(order 000, 100, 010, 110, 001, 101, 011, 111; zeros exact; $p^{(10)}(000, 000) = \frac{1}{4} + \frac{3}{4}3^{-10}$). The rows of $\mathbf{P}^n$ do not converge: they alternate between two limits, while $\pi = (\frac{1}{8}, \dots, \frac{1}{8})$ is stationary all along.

Started at 000, the parity of the number of 1s in X_n equals the parity of n ; π exists, but X_n does not converge in distribution.

- **Cycle** of length L : $d = 2$ if L is even, $d = 1$ if L is odd (round trips of lengths 2 and L).
Ehrenfest: $d = 2$. **Weather, vowel/consonant**: $p(i, i) > 0$, so $d = 1$.
- The lazy chain $\mathbf{P}' = \frac{1}{2}(\mathbf{I} + \mathbf{P})$ has the same π and period 1, so it removes periodicity.



Blue: even parity; amber: odd. Every edge joins the two colours, so the walk alternates colours forever.

Recurrence and Transience

Definition

The **return time** to i is $T_i: \Omega \rightarrow \{1, 2, \dots\} \cup \{\infty\}$, $T_i = \min\{n \geq 1 : X_n = i\}$ (∞ if there is no such n). State i is **recurrent** if $\mathbb{P}_i(T_i < \infty) = 1$ and **transient** if $\mathbb{P}_i(T_i < \infty) < 1$. A recurrent state is **positive recurrent** if $\mathbb{E}_i[T_i] < \infty$ and **null recurrent** otherwise.

A state is recurrent if the chain started there returns with probability one, and transient if with positive probability it never returns.

- T_i is a row quantity, the first time after 0 at which the path is in state i . It is a waiting time of the kind studied in the Conditional Expectation lecture; for the gambler, the duration T of the game was such a time.
- Gambler's ruin: 0 and N are recurrent ($T_i = 1$ surely); $1, \dots, N - 1$ are transient, since the game ends with probability one and the chain then never returns.
- Random walk on $\mathbb{Z}$: state 0 is recurrent iff $p = \frac{1}{2}$ (slide 62). For $p = \frac{1}{2}$, 0 is recurrent but $\mathbb{E}_0[T_0] = \infty$, so the fair walk is null recurrent; this is the statement "ruin certain but $\mathbb{E}[T] = \infty$ " of slide 35.
- For a positive recurrent state, returns occur every $\mathbb{E}_i[T_i]$ steps on average, so the long-run fraction of time spent at i is $1/\mathbb{E}_i[T_i]$ (slide 74).

Recurrence: The Visits Criterion

Theorem

Let $V_i: \Omega \rightarrow \{0, 1, \dots\} \cup \{\infty\}$, $V_i = \sum_{n \geq 1} \mathbf{1}\{X_n = i\}$, be the number of returns to i , and $f = \mathbb{P}_i(T_i < \infty)$. Then

$$\mathbb{P}_i(V_i \geq k) = f^k, \quad \mathbb{E}_i[V_i] = \sum_{n \geq 1} p^{(n)}(i, i) = \frac{f}{1-f} \quad (= \infty \text{ if } f = 1).$$

Hence i is recurrent iff $\sum_n p^{(n)}(i, i) = \infty$ iff $\mathbb{P}_i(V_i = \infty) = 1$; transient iff $\sum_n p^{(n)}(i, i) < \infty$ iff $V_i < \infty$ with probability one.

The number of returns to i is geometric with parameter f , so it is infinite with probability one in the recurrent case and has finite mean in the transient case.

Proof

Each time the chain is at i it restarts (Markov property), and returns once more with probability f , independently of the past; so $V_i \geq k$ requires k successive returns: f^k . Then

$\mathbb{E}_i[V_i] = \sum_{k \geq 1} \mathbb{P}_i(V_i \geq k) = \sum_k f^k$, and also $\mathbb{E}_i[V_i] = \sum_n \mathbb{E}_i[\mathbf{1}\{X_n = i\}] = \sum_n p^{(n)}(i, i)$ by linearity, column by column. □

Recurrence: Class Property and Finite Chains

Theorem

- (a) Recurrence and transience are class properties: all states of a class are recurrent, or all are transient.
- (b) Every finite closed class is recurrent; every class that is not closed is transient. In particular a finite irreducible chain is recurrent, indeed positive recurrent, and its stationary distribution is unique with $\pi_i = 1/\mathbb{E}_i[T_i]$.

In a finite chain a state is transient only if its class can be left permanently; within a finite closed class every state is revisited, on average every $1/\pi_i$ steps.

Proof sketch

(a) If $i \leftrightarrow j$ with $p^{(a)}(i, j)p^{(b)}(j, i) > 0$, then $p^{(a+n+b)}(i, i) \geq p^{(a)}(i, j)p^{(n)}(j, j)p^{(b)}(j, i)$, so the two series $\sum_n p^{(n)}(i, i)$, $\sum_n p^{(n)}(j, j)$ diverge together. (b) Started in a finite closed class, some state is visited infinitely often, so that state is recurrent, so all are by (a). A class that is not closed can be left permanently. The formula $\pi_i = 1/\mathbb{E}_i[T_i]$ is the ergodic theorem, slide 74. $\square$

- Weather: mean return time to R is $1/\pi_R = \frac{13}{3}$ days. Vowel/consonant: mean return to V is $1/\pi_V = \frac{13}{5}$, in agreement with first-step analysis: from C the wait for V is geometric with mean 2, giving $1 + 0.8 \cdot 2$.

The Random Walk on $\mathbb{Z}$ Is Recurrent If and Only If It Is Fair

Theorem

The simple random walk on $\mathbb{Z}$ is recurrent if $p = \frac{1}{2}$ (null recurrent) and transient if $p \neq \frac{1}{2}$.

Without drift the walk returns to 0 infinitely often, with infinite mean return time; with drift it returns only finitely often.

Proof

Returns to 0 take an even number of steps, n up and n down: $p^{(2n)}(0,0) = \binom{2n}{n} p^n q^n$ (slide 7). By Stirling's formula (not proved here) $\binom{2n}{n} \approx 4^n / \sqrt{\pi n}$, the ratio tending to 1, so

$$p^{(2n)}(0,0) \approx \frac{(4pq)^n}{\sqrt{\pi n}}.$$

For $p = \frac{1}{2}$, $4pq = 1$ and $\sum_n (\pi n)^{-1/2} = \infty$, so 0 is recurrent by the visits criterion. For $p \neq \frac{1}{2}$, $4pq = 1 - (p - q)^2 < 1$ and the series converges, so 0 is transient. Null recurrence: $\mathbb{E}_0[T_0] = \infty$ by slide 35. □

- $p = 0.6$: $4pq = 0.96$ and $\mathbb{E}_0[V_0] = \sum_{n \geq 1} \binom{2n}{n} (pq)^n = (1 - 4pq)^{-1/2} - 1 = 4$, so $f = \frac{4}{5} = 1 - |p - q|$ by the visits criterion; the walk returns four times on average.

Hitting Probabilities: The Minimal Solution of a First-Step System

Definition

For $A \subseteq S$, the **hitting time** of A is $T_A: \Omega \rightarrow \{0, 1, 2, \dots\} \cup \{\infty\}$, $T_A = \min\{n \geq 0 : X_n \in A\}$, and the **hitting probability** is the function

$$h^A: S \rightarrow [0, 1], \quad h_i^A = \mathbb{P}_i(T_A < \infty).$$

Theorem

h^A is the **minimal non-negative solution** of the system

$$h_i = 1 \quad (i \in A), \quad h_i = \sum_{j \in S} p(i, j) h_j \quad (i \notin A) :$$

it solves it, and every solution x with all $x_i \geq 0$ has $x_i \geq h_i^A$ for all i .

Off the target set, the hitting probability from i is the $p(i, \cdot)$ -weighted average of the hitting probabilities from the next states; if the system has several non-negative solutions, the hitting probability is the smallest.

- A is the fixed target set; i is the starting state, the variable of h^A . Gambler's ruin: $S = \{0, \dots, N\}$, $A = \{N\}$ (next slide); the exit problem of slide 11: $A = \{-a, a\}$.

Hitting Probabilities: Proof

Proof

Solves it: $h_i^A = 1$ on A ; off A , condition on the first step (total probability over the rows $\{X_1 = j\}$) and use the Markov property: $\mathbb{P}_i(T_A < \infty) = \sum_j p(i, j) \mathbb{P}_j(T_A < \infty)$. Minimal: let $x \geq 0$ solve the system. Then $x_i \geq \mathbb{P}_i(T_A \leq n)$ for all n , by induction: true for $n = 0$ (on A , $x_i = 1$; off A , $x_i \geq 0 = \mathbb{P}_i(T_A \leq 0)$); and for $i \notin A$,

$$x_i = \sum_j p(i, j) x_j \geq \sum_j p(i, j) \mathbb{P}_j(T_A \leq n) = \mathbb{P}_i(T_A \leq n + 1)$$

by the induction hypothesis and first-step analysis. Let $n \rightarrow \infty$: $x_i \geq \mathbb{P}_i(T_A < \infty) = h_i^A$. □

Any non-negative solution dominates the probability of hitting within n steps for every n , hence dominates the probability of ever hitting.

- On a finite state space in which A can be reached from every state, the system has a unique solution and minimality plays no role; when there are absorbing states outside A , or the state space is infinite, minimality selects the correct solution.

Hitting Probabilities: The Gambler's Ruin Is the Case $A = \{N\}$

Finite ruin chain, $A = \{N\}$

The system reads $h_N = 1$; $h_i = p h_{i+1} + q h_{i-1}$ for $0 < i < N$; and at the absorbing state 0: $h_0 = \sum_j p(0, j) h_j = h_0$, which imposes no constraint. The general solution is $h_i = a + b r^i$ (or $a + bi$ if $p = q$) for $i \geq 1$ with h_0 free; non-negative solutions exist with any $h_0 \in [0, 1]$, and the minimal one has $h_0 = 0$, which recovers slides 29–30.

The boundary condition $h_0 = 0$ imposed on slide 28 is supplied here by minimality; an absorbing state has hitting probability 0 for any target set not containing it.

- The hitting probabilities of the ruin problem come from this system with different targets: $A = \{N\}$ gives the probability of reaching the target, $A = \{0\}$ the probability of ruin, and $A = \{0, N\}$ gives $h \equiv 1$, so the game ends with probability one.
- The ruin argument of slide 28, conditioning on the first round and restarting, is the proof that h^A solves the system; the only addition for general chains is minimality.

Why “Minimal”: The Infinite Ruin Chain

Infinite ruin chain, $S = \{0, 1, 2, \dots\}$, $A = \{0\}$ (slide 34 again)

$h_0 = 1$ and $h_i = p h_{i+1} + q h_{i-1}$ for $i \geq 1$; general solution $h_i = 1 + b(r^i - 1)$ with $r = q/p$.

- $p > q$ ($r < 1$): $h_i \geq 0$ for all i requires $b \leq 1$, and h_i decreases as b increases, so the minimal solution is $b = 1$: $h_i = r^i = (q/p)^i$.
- $p < q$ ($r > 1$): $h_i \geq 0$ for all i requires $b \geq 0$ (otherwise $h_i \rightarrow -\infty$), minimal at $b = 0$: $h_i = 1$.
- $p = q$: $h_i = 1 + bi$, $b \geq 0$, minimal $h_i = 1$.

$h \equiv 1$ solves the system in every case; it is the answer only when it is the smallest non-negative solution.

The system alone does not distinguish certain ruin from ruin with probability $(q/p)^i$; the minimality condition does, and the result agrees with the limit $N \rightarrow \infty$ on slide 34.

- The same occurs for transient chains in general: on an infinite state space, a non-negative solution can be inflated by adding a positive solution of the homogeneous equation that vanishes on A ; only the smallest is the hitting probability.

Expected Hitting Times

Theorem

Let $k^A: S \rightarrow [0, \infty]$, $k_i^A = \mathbb{E}_i[T_A]$. Then k^A is the minimal non-negative solution of

$$k_i = 0 \quad (i \in A), \quad k_i = 1 + \sum_{j \in S} p(i, j) k_j \quad (i \notin A).$$

Off the target set, the expected hitting time is one step plus the weighted average of the expected hitting times from the next states; the exit-time and duration equations of slides 11 and 31 are instances of this system.

Proof

Solves it: off A , $T_A = 1 + T'_A$ where T'_A is the hitting time of the chain restarted at X_1 ; take $\mathbb{E}_i$ and condition on X_1 . Minimal: as before, any non-negative solution x satisfies $x_i \geq \mathbb{E}_i[\min(T_A, n)]$ for all n , by induction. $\square$

- The system may have no finite non-negative solution; then $k_i^A = \infty$ for some i , as for the fair walk on $\mathbb{Z}$. A finite solution, when the target is reachable from everywhere in a finite chain, is unique.
- In matrix form on a finite chain: $(\mathbf{I} - \mathbf{Q})k = \mathbf{1}$ with $\mathbf{Q}$ the transitions among the states off A ; slide 77 inverts it.

Expected Hitting Times: Corner to Opposite Corner on the Cube

We compute the expected number of steps for the random walk on the cube to go from 000 to 111.

Direct computation

$\mathbb{E}_{000}[T_{111}] = \sum_n n \mathbb{P}_{000}(T_{111} = n)$, where each $\mathbb{P}_{000}(T_{111} = n)$ is a sum over paths of length n that avoid 111 until their last step; the quantity is not determined by the distribution of X_n at any fixed time.

Random walk on the cube: from 000 to the opposite corner 111

Group states by distance $d \in \{0, 1, 2, 3\}$ from 111; by symmetry k depends only on d . From distance d a step goes to distance $d - 1$ with probability $d/3$ and to $d + 1$ with probability $(3 - d)/3$:

$$k_0 = 0, \quad k_1 = 1 + \frac{1}{3}k_0 + \frac{2}{3}k_2, \quad k_2 = 1 + \frac{2}{3}k_1 + \frac{1}{3}k_3, \quad k_3 = 1 + k_2,$$

giving $k_1 = 7$, $k_2 = 9$, $k_3 = 10$: ten steps on average from a corner to the opposite corner.

Symmetry reduces the eight unknowns to three, and first-step analysis replaces the infinite sum over paths by three linear equations.

- Check: from a neighbour of 000 the expected time back to 000 is 7, so the mean return time to 000 is $1 + 7 = 8 = 1/\pi_{000}$, as the ergodic theorem predicts (slide 74).

Return to Question 3: Days Until Rain

$A = \{R\}$, $k_R = 0$, and for the other two states

$$k_S = 1 + 0.7 k_S + 0.2 k_C, \quad k_C = 1 + 0.3 k_S + 0.4 k_C.$$

Answer

From the second, $0.6 k_C = 1 + 0.3 k_S$; substituting into the first, $0.3 k_S = 1 + 0.2 k_C$, so $k_S = \frac{20}{3} \approx 6.7$ days and $k_C = 5$ days.

Consistency check with the stationary distribution

The mean return time to R is one step plus the expected hitting time from wherever that step lands:

$$\mathbb{E}_R[T_R] = 1 + 0.2 k_S + 0.4 k_C + 0.4 \cdot 0 = 1 + \frac{4}{3} + 2 = \frac{13}{3} = \frac{1}{\pi_R}.$$

The three opening questions have been answered by a matrix power, a left eigenvector, and a first-step linear system respectively.

- For the vowel/consonant chain, the expected number of letters until the first vowel is 2 from C (a geometric waiting time) and $1 + 0.8 \cdot 2 = 2.6$ from V; until the first occurrence of VV it is $k_C = 15$ and $k_V = 13$, from the first-step system $k_C = 2 + k_V$, $k_V = 1 + 0.8 k_C$.

The Convergence Theorem

Theorem

Let $\mathbf{P}$ be irreducible and aperiodic on a finite state space S . Then $\mathbf{P}$ has a unique stationary distribution π , and for all $i, j \in S$,

$$p^{(n)}(i, j) \xrightarrow{n \rightarrow \infty} \pi_j.$$

Equivalently, for every initial distribution λ , $\lambda \mathbf{P}^n \rightarrow \pi$: every row of $\mathbf{P}^n$ converges to π .

For large n the distribution of X_n is close to π , whatever the initial distribution.

- Slide 48: rows of the vowel/consonant $\mathbf{P}^n$ approaching $(\frac{5}{13}, \frac{8}{13})$. Slide 47: rows of the weather $\mathbf{P}^7$ agreeing to two places. PageRank: $\mathbf{P}^{30}$ has all rows equal to π to four places, which is how π is computed for 10^{10} pages.
- The theorem concerns the *columns* of the table of paths, the distribution at a fixed late time. It does not describe an individual path, which continues to move; the ergodic theorem (slide 74) is the corresponding statement about rows.
- Both hypotheses are necessary: irreducibility for the uniqueness of π and the independence of the limit from the start, aperiodicity for the convergence itself. Examples follow on the next slide.

Why Each Hypothesis Is Needed

Reducible: the limit depends on the start

Two closed classes, $\mathbf{P} = \begin{pmatrix} \mathbf{P}_1 & 0 \\ 0 & \mathbf{P}_2 \end{pmatrix}$. Rows of $\mathbf{P}^n$ starting in class 1 converge to $(\pi_1, 0)$, in class 2 to $(0, \pi_2)$; every mixture $(\alpha\pi_1, (1-\alpha)\pi_2)$ is stationary. Gambler's ruin: rows converge to $(1 - h_i, 0, \dots, 0, h_i)$, different for each i ; for $N = 4$, $p = \frac{1}{2}$, row i tends to $(1 - \frac{i}{4}, 0, 0, 0, \frac{i}{4})$.

Periodic: the rows do not converge

The cube (slide 58): rows of $\mathbf{P}^n$ alternate between two vectors, though $\pi = (\frac{1}{8}, \dots, \frac{1}{8})$ is unique. Likewise the Ehrenfest urn and the cycle of even length.

Infinite S : no stationary distribution

The fair walk on $\mathbb{Z}$ is irreducible, but has no stationary distribution (slide 50) and $p^{(2n)}(0, 0) \approx 1/\sqrt{\pi n} \rightarrow 0$ (slide 62): every entry of every row of $\mathbf{P}^n$ tends to 0, and the mass spreads out over ever more states.

Without irreducibility the limit depends on the starting state, without aperiodicity there may be no limit, and on an infinite state space the limit may be zero at every state.

How Fast: The Second Eigenvalue

Theorem (rate of convergence, diagonalisable case)

Suppose $\mathbf{P}$ is diagonalisable, with eigenvalues $1 = \lambda_1, \lambda_2, \dots, \lambda_{|S|}$, right eigenvectors $\mathbf{v}_k$ (columns, $\mathbf{P}\mathbf{v}_k = \lambda_k\mathbf{v}_k$) and left eigenvectors $\mathbf{u}_k$ (rows, $\mathbf{u}_k\mathbf{P} = \lambda_k\mathbf{u}_k$), chosen so that $\mathbf{u}_k\mathbf{v}_k = 1$ and $\mathbf{u}_k\mathbf{v}_l = 0$ for $k \neq l$. Then $\mathbf{v}_1 = \mathbf{1}$, $\mathbf{u}_1 = \pi$, $\mathbf{P} = \sum_k \lambda_k \mathbf{v}_k \mathbf{u}_k$, and

$$\mathbf{P}^n = \mathbf{1}\pi + \sum_{k \geq 2} \lambda_k^n \mathbf{v}_k \mathbf{u}_k, \quad |p^{(n)}(i, j) - \pi_j| \leq C |\lambda_2|^n, \quad |\lambda_2| = \max_{k \geq 2} |\lambda_k| < 1.$$

Each term of $\mathbf{P}^n$ other than $\mathbf{1}\pi$ carries a factor λ_k^n with $|\lambda_k| < 1$, so the difference from the limit decreases geometrically at a rate set by the second-largest eigenvalue modulus.

Proof

$\mathbf{P}\mathbf{1} = \mathbf{1}$ (row sums) and $\pi\mathbf{P} = \pi$, $\pi\mathbf{1} = 1$ give $\mathbf{v}_1 = \mathbf{1}$, $\mathbf{u}_1 = \pi$. As $(\mathbf{v}_k\mathbf{u}_k)(\mathbf{v}_l\mathbf{u}_l) = (\mathbf{u}_k\mathbf{v}_l)\mathbf{v}_k\mathbf{u}_l$ vanishes for $k \neq l$, powers act termwise: $(\sum_k \lambda_k \mathbf{v}_k \mathbf{u}_k)^n = \sum_k \lambda_k^n \mathbf{v}_k \mathbf{u}_k$. Irreducibility rules out a second eigenvalue 1 (a second stationary distribution), aperiodicity the other eigenvalues of modulus 1 (the cube has -1); these two facts are not proved here. $\square$

- Vowel/consonant: $\lambda_2 = -0.3$, error $\frac{8}{13}(0.3)^n$ (slide 48). Weather: eigenvalues 1, 0.456, 0.044; $\max_{i,j} |p^{(n)}(i, j) - \pi_j| = 0.26, 0.056, 0.0053, 0.0024$ at $n = 1, 3, 6, 7$.

Proof Sketch by Coupling

Idea

Run two independent copies of the chain: X started at i , and Y started from π , so that $Y_n \sim \pi$ for every n (slide 49). Let $\tau = \min\{n : X_n = Y_n\}$ be the first time they meet. Define $Z_n = X_n$ for $n < \tau$ and $Z_n = Y_n$ for $n \geq \tau$: switch to following Y once they coincide.

Two facts

1. Z is again a Markov chain with matrix $\mathbf{P}$ started at i (at the switch, both copies are at the same state and both continue by the same rule), so $\mathbb{P}(Z_n = j) = p^{(n)}(i, j)$.
2. $Z_n = Y_n$ whenever $\tau \leq n$, so $|p^{(n)}(i, j) - \pi_j| = |\mathbb{P}(Z_n = j) - \mathbb{P}(Y_n = j)| \leq \mathbb{P}(\tau > n)$.

Where the hypotheses enter

The pair (X_n, Y_n) is a Markov chain on $S \times S$ with transitions $p(i, i')p(j, j')$. Irreducibility and aperiodicity of $\mathbf{P}$ make this pair chain irreducible (aperiodicity is needed: for the cube, a pair at different parities can never meet), hence recurrent on the finite $S \times S$, so it reaches the diagonal: $\mathbb{P}(\tau < \infty) = 1$, i.e. $\mathbb{P}(\tau > n) \rightarrow 0$. □

After the two chains first meet, the chain started at i can be replaced by the chain started from π without changing its distribution, and the two are certain to meet.

The Ergodic Theorem: Long-Run Fractions of Time

Theorem

Let $\mathbf{P}$ be irreducible on a finite S with stationary distribution π (no aperiodicity required). Let $V_i(n): \Omega \rightarrow \{0, \dots, n\}$, $V_i(n) = \sum_{k=0}^{n-1} \mathbf{1}\{X_k = i\}$, be the number of visits to i before time n . Unlike V_i of slide 60, this count includes time 0, so $V_i(n) \rightarrow V_i + \mathbf{1}\{X_0 = i\}$. Then, whatever the initial distribution, with probability one

$$\frac{V_i(n)}{n} \xrightarrow{n \rightarrow \infty} \pi_i \quad \text{for every } i, \quad \text{and} \quad \frac{1}{n} \sum_{k=0}^{n-1} f(X_k) \rightarrow \sum_{i \in S} \pi_i f(i) \quad \text{for every } f: S \rightarrow \mathbb{R}.$$

Moreover $\pi_i = 1/\mathbb{E}_i[T_i]$.

Along a single path, the fraction of time spent in state i converges to π_i , and the time average of any function of the state converges to its average under π .

- $V_i(n)/n$ is a *row* quantity, computed by counting along one path. The theorem says that almost every row has the same limiting frequencies, namely π .
- f is a reward per state: long-run average reward equals the π -weighted reward. With $f = \mathbf{1}_{\{i\}}$ this is the first statement.

The Ergodic Theorem: Proof Idea and a Time Average

Proof idea

The visits to i split the path into independent, identically distributed excursions of mean length $\mathbb{E}_i[T_i]$ (Markov property at each return). By the law of large numbers (LLN: averages of i.i.d. variables with finite mean converge to that mean with probability one), m excursions take about $m \mathbb{E}_i[T_i]$ steps, so in n steps there are about $n/\mathbb{E}_i[T_i]$ visits: $V_i(n)/n \rightarrow \mu_i := 1/\mathbb{E}_i[T_i]$.

Why the limit is π

Among the visits to i , a fraction $p(i, j)$ is followed by a step to j (Markov property, LLN), so the fraction of times $k < n$ with $X_k = i$, $X_{k+1} = j$ tends to $\mu_i p(i, j)$. Summing over i counts the visits to j at times $1, \dots, n$, whose fraction tends to μ_j : $\mu_j = \sum_i \mu_i p(i, j)$, i.e. $\mu \mathbf{P} = \mu$, so $\mu = \pi$ by uniqueness (slide 61). □

The frequencies satisfy $\mu \mathbf{P} = \mu$ because every arrival at j is a departure from some i .

The cube: a time average although the rows of $\mathbf{P}^n$ never converge

$f(x)$ = number of 1s in x . Along any path, with probability one,
 $\frac{1}{n} \sum_{k < n} f(X_k) \rightarrow \sum_i \pi_i f(i) = \frac{1}{8}(0 + 3 \cdot 1 + 3 \cdot 2 + 3) = \frac{3}{2}$, and the fraction of time at even-parity corners tends to $\frac{1}{2}$, although at each fixed time n that probability is 1 or 0 (slide 58).

Return to Question 2, and the Two Theorems Side by Side

Weather, question 2

In the long run a fraction $\pi_S = \frac{6}{13} \approx 0.46$ of days are sunny, $\frac{4}{13}$ cloudy, $\frac{3}{13}$ rainy: with probability one, along the actual sequence of days. And the fraction of days that are rainy *and* followed by a sunny day is $\pi_R p(R, S) = \frac{3}{13} \cdot 0.2 = \frac{3}{65} \approx 0.046$ (apply the theorem to the pair chain (X_n, X_{n+1})).

	convergence theorem	ergodic theorem
statement	$\mathbb{P}(X_n = i) \rightarrow \pi_i$	$V_i(n)/n \rightarrow \pi_i$ with probability one
reads the table	down a column (late time)	along a row (one path)
needs	irreducible, aperiodic, finite	irreducible, finite
the cube ($d = 2$)	does not apply: rows of $\mathbf{P}^n$ alternate	applies: $\frac{1}{8}$ of the time at each corner
what π_i means	distribution of X_n for large n	long-run fraction of time at i
computes	$\mathbf{P}^n$ for large n	π from $\pi \mathbf{P} = \pi$, or $1/\mathbb{E}_i[T_i]$

Both theorems describe the long-run behaviour of the chain in terms of π : the first as the distribution at a fixed late time, the second as the frequencies along a single path.

Absorbing Chains and the Fundamental Matrix

Definition

A chain is **absorbing** if every state leads to some absorbing state. Order the states transient first, absorbing last: with S_T the transient and S_A the absorbing states,

$$\mathbf{P} = \begin{pmatrix} \mathbf{Q} & \mathbf{R} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}, \quad \mathbf{Q}: S_T \times S_T \rightarrow [0, 1], \quad \mathbf{R}: S_T \times S_A \rightarrow [0, 1].$$

The **fundamental matrix** is $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}: S_T \times S_T \rightarrow [0, \infty)$.

Theorem

$\mathbf{I} - \mathbf{Q}$ is invertible and $\mathbf{N} = \sum_{n \geq 0} \mathbf{Q}^n$. For transient i, j and absorbing a :

$$\mathbf{N}_{ij} = \mathbb{E}_i[\#\{n \geq 0 : X_n = j\}], \quad (\mathbf{N}\mathbf{1})_i = \mathbb{E}_i[T_{S_A}], \quad (\mathbf{NR})_{ia} = \mathbb{P}_i(\text{absorbed at } a).$$

$\mathbf{Q}^n$ contains the n -step transition probabilities among the transient states, so its sum over n gives expected numbers of visits; the row sums of $\mathbf{N}$ are expected times to absorption, and $\mathbf{NR}$ gives the absorption probabilities.

The Fundamental Matrix: Proof Sketch

Proof sketch

$(Q^n)_{ij} = \mathbb{P}_i(X_n = j)$ for transient j (paths through transient states only), so $\sum_n (Q^n)_{ij} = \mathbb{E}_i[\text{visits to } j]$ by linearity, column by column. Absorption is certain and from a finite set of transient states happens geometrically fast, so $Q^n \rightarrow 0$ and the series converges, to $(I - Q)^{-1}$ since $(I - Q) \sum_{n < m} Q^n = I - Q^m$. Time to absorption is the total number of transient visits: $\sum_j N_{ij}$. Absorption at a occurs by a visit to some transient j followed by a step to a , giving $\sum_j N_{ij} R_{ja}$. $\square$

The expected numbers of visits, the expected time to absorption and the absorption probabilities are all obtained from the single matrix $(I - Q)^{-1}$.

- The hitting systems of slides 63 and 67 are $(I - Q)k = \mathbf{1}$ and $(I - Q)h = \mathbf{R}_{\cdot, a}$; the fundamental matrix is their solution operator.
- For a hitting question, any chain can be made absorbing by making the target states absorbing (replacing their rows by point masses), which does not change the path up to the hitting time.

The Fundamental Matrix of the Ruin Chain, $N = 4$

Transient states 1, 2, 3; absorbing 0, 4. Fair case:

$$\mathbf{Q} = \begin{pmatrix} 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \quad \mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1} = \begin{pmatrix} \frac{3}{2} & 1 & \frac{1}{2} \\ 1 & 2 & 1 \\ \frac{1}{2} & 1 & \frac{3}{2} \end{pmatrix},$$

$$\mathbf{N}\mathbf{1} = \begin{pmatrix} 3 \\ 4 \\ 3 \end{pmatrix}, \quad \mathbf{N}\mathbf{R} = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}.$$

Reading it

$\mathbf{N}\mathbf{1} = (3, 4, 3) = i(4 - i)$: the durations of slide 31. Column 4 of $\mathbf{N}\mathbf{R}$ is $(\frac{1}{4}, \frac{1}{2}, \frac{3}{4}) = i/4$: slide 29. The matrix also gives new information: $\mathbf{N}_{2,2} = 2$ is the expected number of visits to fortune 2 (counting the start) before the game ends when starting there, and $\mathbf{N}_{1,3} = \frac{1}{2}$ the expected number of visits to 3 when starting from 1.

- $p = \frac{2}{5}$: $\mathbf{N}\mathbf{1} = (\frac{33}{13}, \frac{50}{13}, \frac{43}{13})$ and the target column of $\mathbf{N}\mathbf{R}$ is $(\frac{8}{65}, \frac{4}{13}, \frac{38}{65})$, agreeing with the closed forms of slides 30–31. The single matrix inverse gives both quantities at once.

Summary

object	type	meaning in the table of paths
process $(X_t)_{t \in T}$	$X_t: \Omega \rightarrow \mathbb{R}$ for each t	columns are random variables, rows are paths
random walk S_n	$\Omega \rightarrow \mathbb{Z}$	sum of ± 1 steps; $\text{Var}(S_n) = 4npq$ ($= n$ if fair); restarts after each step
Brownian motion B_t	$[0, \infty) \times \Omega \rightarrow \mathbb{R}$	the walk at $\sqrt{n}$ scale; independent $N(0, t - s)$ increments
ruin probability $1 - h_i$	$\{0, \dots, N\} \rightarrow [0, 1]$	$1 - i/N$, or $\frac{r^N - r^i}{r^N - 1}$; from $h_i = ph_{i+1} + qh_{i-1}$
transition matrix $\mathbf{P}$	$S \times S \rightarrow [0, 1]$, rows sum to 1	row i is the conditional pmf of X_{n+1} given $X_n = i$
n -step $\mathbf{P}^n$, law $\lambda \mathbf{P}^n$	matrix, row vector	Chapman–Kolmogorov: condition on the middle state
stationary π	$S \rightarrow [0, 1]$, $\pi \mathbf{P} = \pi$	inflow equals outflow at every state; detailed balance if reversible
classes, period, recurrence	structure of $\mathbf{P}$'s graph	which states reach which, at what step lengths, and whether return is certain
hitting h^A , k^A	$S \rightarrow [0, 1]$, $S \rightarrow [0, \infty]$	minimal solutions of first-step systems; NR , N1
convergence, ergodic	$\mathbf{P}^n \rightarrow \mathbf{1}\pi$; $V_i(n)/n \rightarrow \pi_i$	the column limit and the row limit are both π

Most results follow from conditioning on the first step, turning a question about unbounded paths into a finite linear system.